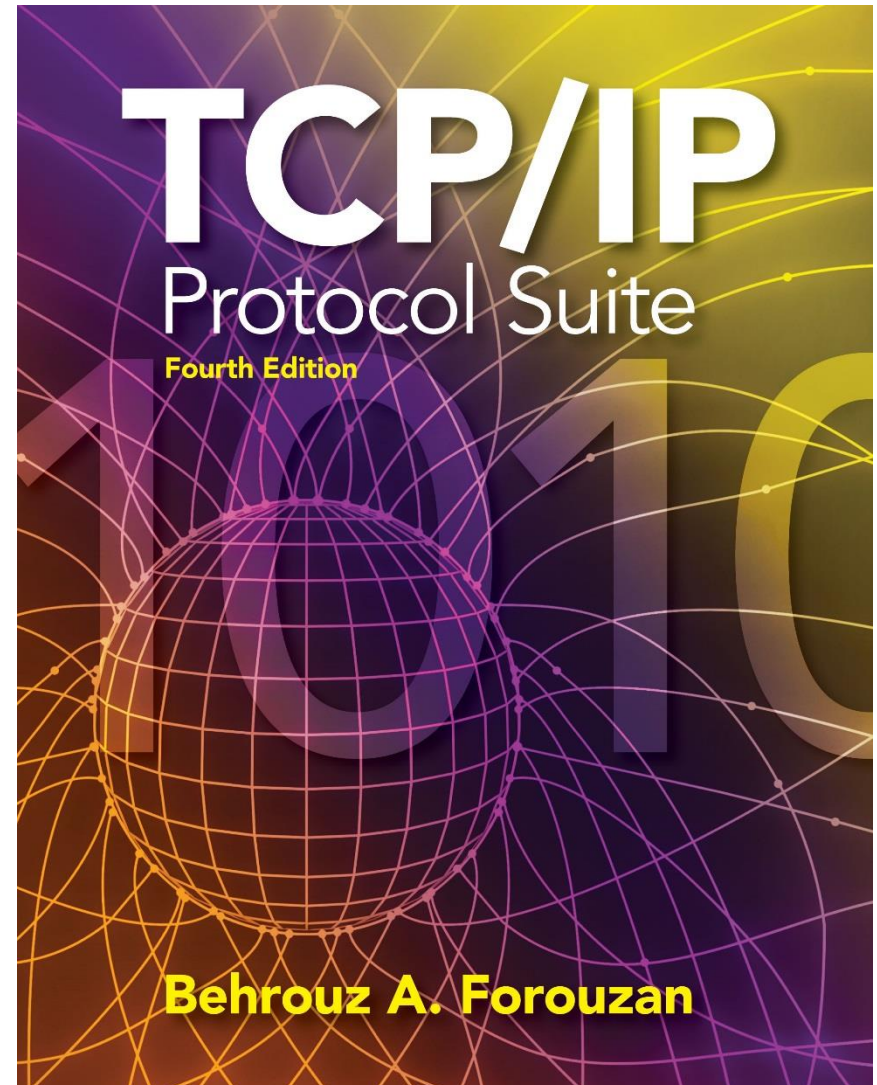


Chapter 12

Multicasting And Multicast Routing Protocols



OBJECTIVES:

- ☐ To compare and contrast unicasting, multicasting, and broadcasting communication.
- ☐ To define multicast addressing space in IPv4 and show the division of the space into several blocks.
- ☐ To discuss the IGMP protocol, which is responsible for collecting group membership information in a network.
- ☐ To discuss the general idea behind multicast routing protocols and their division into two categories based on the creation of the shortest path trees.
- ☐ To discuss multicast link state routing in general and its implementation in the Internet: a protocol named MOSPF.

OBJECTIVES *(continued)*:

- ❑ To discuss multicast distance vector routing in general and its implementation in the Internet: a protocol named DVMRP.
- ❑ To discuss core-based protocol (CBT) and briefly discuss two independent multicast protocols PIM-DM and PIM-SM.
- ❑ To discuss multicast backbone (MBONE) that shows how to create a tunnel when the multicast messages need to pass through an area with no multicast routers.

Chapter Outline

12.1 Introduction

12.2 Multicast Addresses

12.3 IGMP

12.4 Multicast Routing

12.5 Routing Protocols

12.6 MBONE

12-1 INTRODUCTION

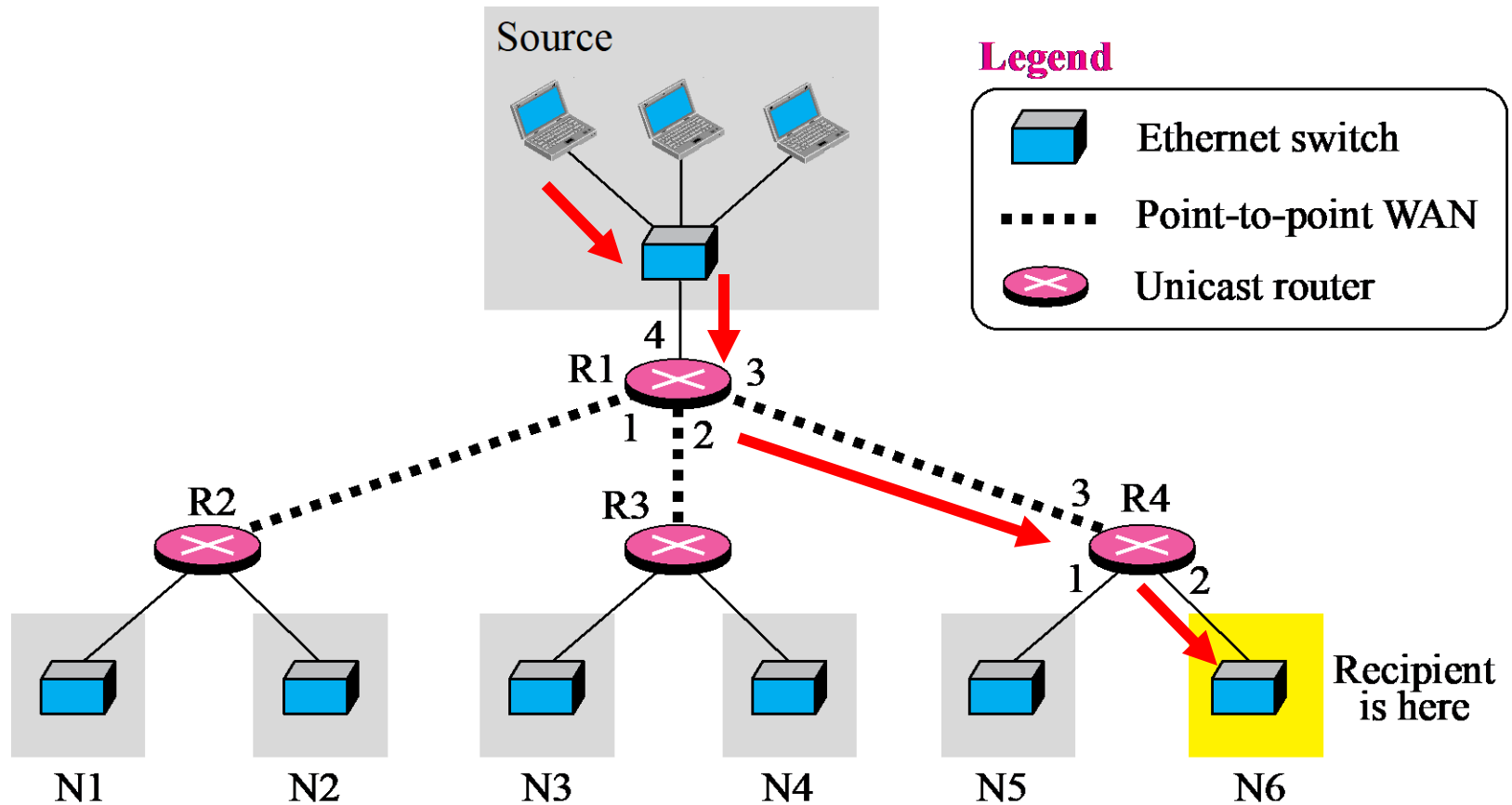
We have learned that forwarding a datagram is normally based on the prefix of the destination address in the datagram. Address aggregation mechanism may combine several datagrams to be delivered to an ISP and then separate them to be delivered to their final destination networks, but the principle does not change.

Understanding the above forwarding principle, we can now define unicasting, multicasting, and broadcasting. Let us clarify these terms as they relate to the Internet.

Topics Discussed in the Section

- ✓ **Unicasting**
- ✓ **Multicasting**
- ✓ **Broadcasting**

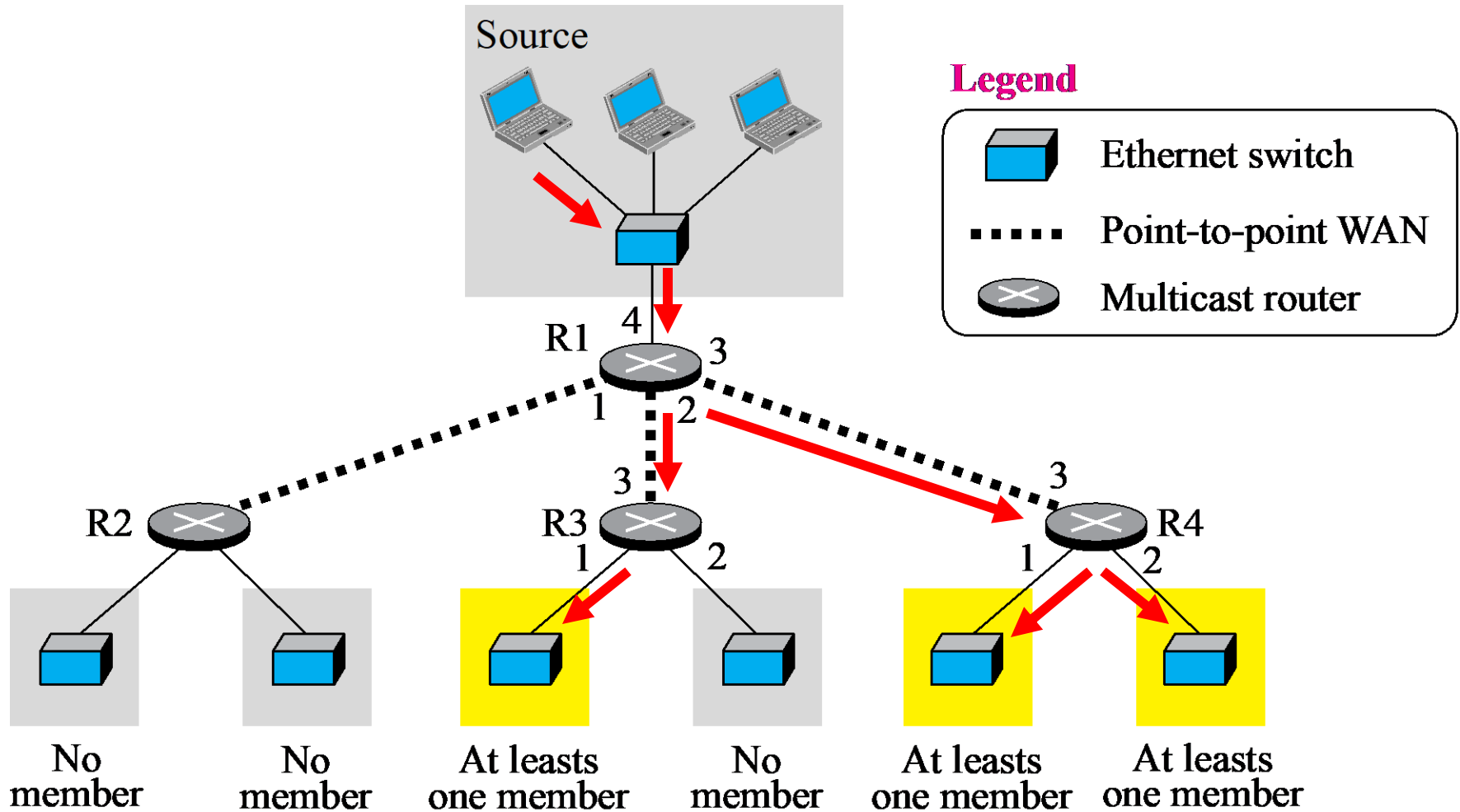
Figure 12.1 *Unicasting*



Note

In unicasting, the router forwards the received datagram through only one of its interfaces.

Figure 12.2 *Multicasting*



Note

In multicasting, the router may forward the received datagram through several of its interfaces.

Figure 12.3 *Multicasting versus multiple unicasting*

Legend



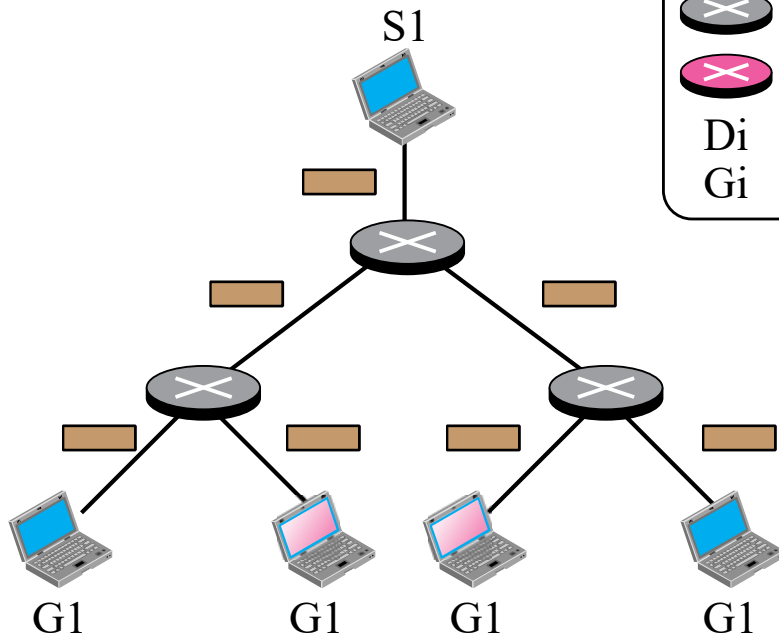
Multicast router



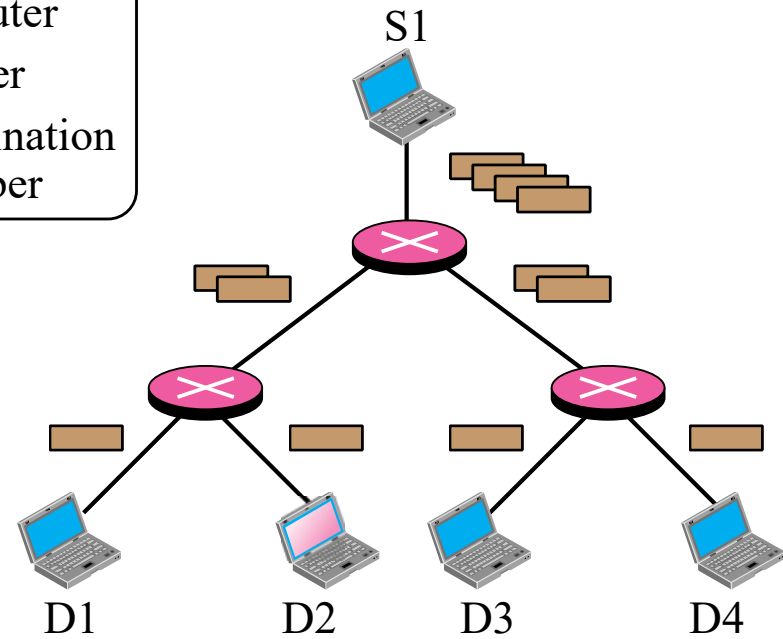
Unicast router

D_i Unicast destination

G_i Group member



a. Multicasting



b. Multiple unicasting

Note

Emulation of multicasting through multiple unicasting is not efficient and may create long delays, particularly with a large group.

12-2 MULTICAST ADDRESSES

A multicast address is a destination address for a group of hosts that have joined a multicast group. A packet that uses a multicast address as a destination can reach all members of the group unless there are some filtering restriction by the receiver.

Topics Discussed in the Section

- ✓ **Multicast Addresses in IPv4**
- ✓ **Selecting Multicast Addresses**
- ✓ **Delivery of Multicast Packets at Data Link Layer**

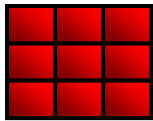


Table 12.1 *Multicast Address Ranges*

<i>CIDR</i>	<i>Range</i>	<i>Assignment</i>
224.0.0.0/24	224.0.0.0 → 224.0.0.255	Local Network Control Block
224.0.1.0/24	224.0.1.0 → 224.0.1.255	Internetwork Control Block
	224.0.2.0 → 224.0.255.255	AD HOC Block
224.1.0.0/16	224.1.0.0 → 224.1.255.255	ST Multicast Group Block
224.2.0.0/16	224.2.0.0 → 224.2.255.255	SDP/SAP Block
	224.3.0.0 → 231.255.255.255	Reserved
232.0.0.0/8	232.0.0.0 → 224.255.255.255	Source Specific Multicast (SSM)
233.0.0.0/8	233.0.0.0 → 233.255.255.255	GLOP Block
	234.0.0.0 → 238.255.255.255	Reserved
239.0.0.0/8	239.0.0.0 → 239.255.255.255	Administratively Scoped Block

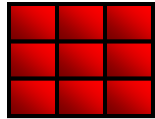


Table 12.2 *Some addresses in Network Control Block*

<i>Address</i>	<i>Assignment</i>
224.0.0.0	Base address (reserved)
224.0.0.1	All systems (hosts or routers) on this network
224.0.0.2	All routers on this network
224.0.0.4	DMVRP routers
224.0.0.5	OSPF routers
224.0.0.7	ST (stream) routers
224.0.0.8	ST (stream) hosts
224.0.0.9	RIP2 routers
224.0.0.10	IGRP routers
224.0.0.11	Mobile Agents
224.0.0.12	DHCP servers
224.0.0.13	PIM routers
224.0.0.14	RSVP encapsulation
224.0.0.15	CBT routers
224.0.0.22	IGMPv3

Example 12.1

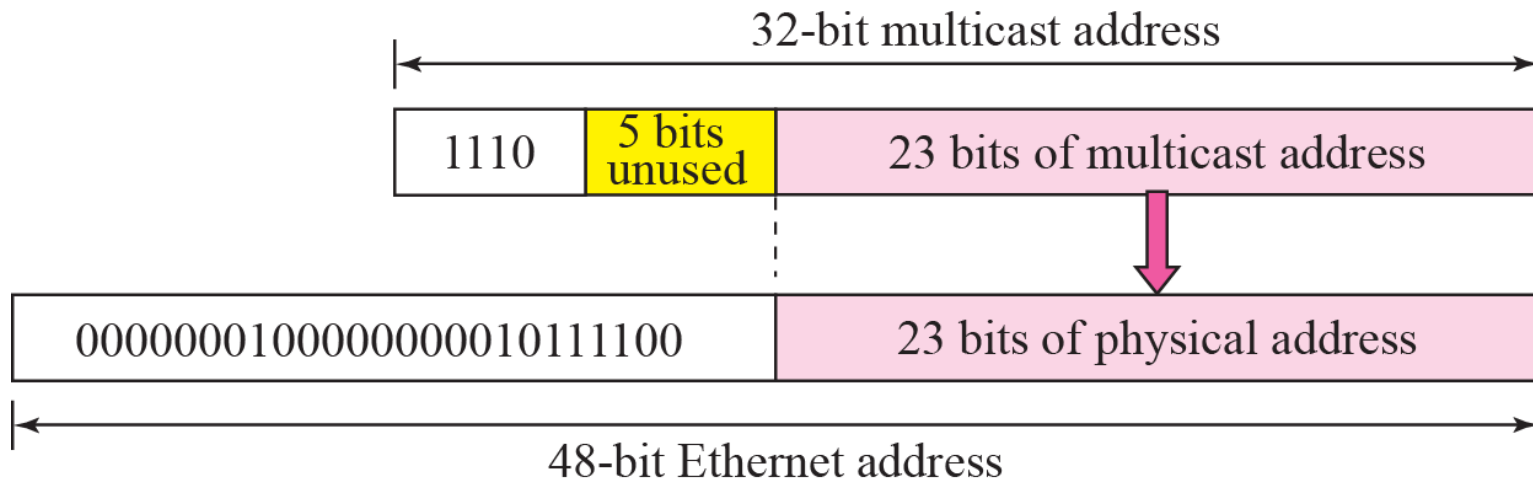
We use `netstat` with three options, `-n`, `-r`, and `-a`. The `-n` option gives the numeric versions of IP addresses, the `-r` option gives the routing table, and the `-a` option gives all addresses (unicast and multicast). Note that we show only the fields relative to our discussion. The multicast address is shown in color.

```
$netstat -nra
```

```
Kernel IP routing table
```

Destination	Gateway	Mask	Flags	Iface
153.18.16.0	0.0.0.0	255.255.240.0	U	eth0
169.254.0.0	0.0.0.0	255.255.0.0	U	eth0
127.0.0.0	0.0.0.0	255.0.0.0	U	lo
224.0.0.0	0.0.0.0	224.0.0.0	U	eth0
0.0.0.0	153.18.31	0.0.0.0	UG	eth0

Figure 12.4 *Mapping class D to Ethernet physical address*



Note

*An Ethernet multicast physical
address is in the range*

01:00:5E:00:00:00

to

01:00:5E:7F:FF:FF.

Example 12.2

Change the multicast IP address 232.43.14.7 to an Ethernet multicast physical address.

Solution

We can do this in two steps:

- a.** We write the rightmost 23 bits of the IP address in hexadecimal. This can be done by changing the rightmost 3 bytes to hexadecimal and then subtracting 8 from the leftmost digit if it is greater than or equal to 8. In our example the result is 2B:0E:07.
- b.** We add the result of part a to the starting Ethernet multicast address, which is 01:00:5E:00:00:00. The result is **01:00:5E:2B:0E:07**

Example 12.3

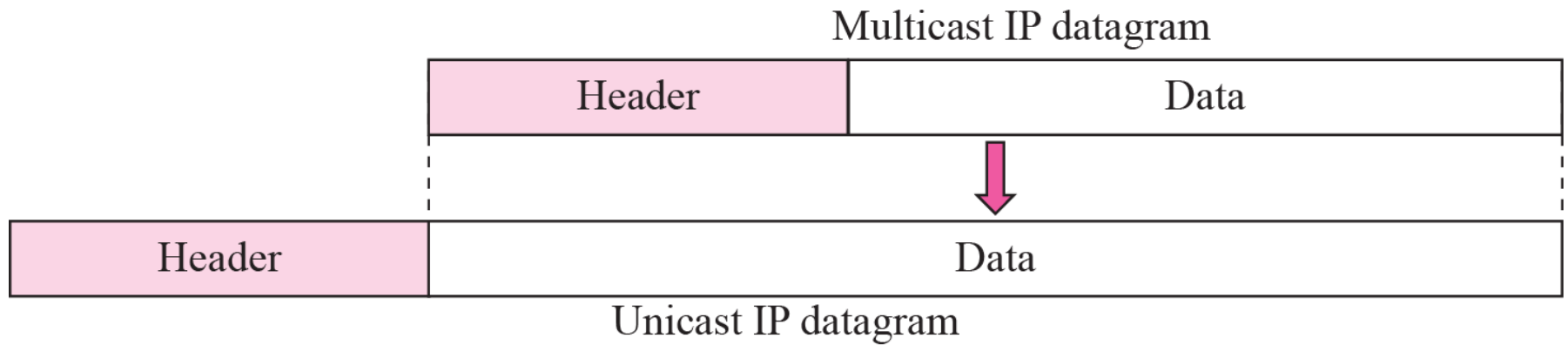
Change the multicast IP address 238.212.24.9 to an Ethernet multicast address.

Solution

We can do this in two steps:

- a. The rightmost 3 bytes in hexadecimal are D4:18:09. We need to subtract 8 from the leftmost digit, resulting in 54:18:09.
- b. We add the result of part a to the Ethernet multicast starting address. The result is **01:00:5E:54:18:09**

Figure 12.5 *Tunneling*



12-3 IGMP

Multicast communication means that a sender sends a message to a group of recipients that are members of the same group. Each multicast router needs to know the list of groups that have at least one loyal member related to each interface. Collection of this type of information is done at two levels: locally and globally. The first task is done by the IGMP protocol; the second task is done by the multicast routing protocols.

Topics Discussed in the Section

- ✓ **Group Management**
- ✓ **IGMP Messages**
- ✓ **IGMP Protocol Applied to host**
- ✓ **IGMP Protocol Applied to Router**
- ✓ **Role of IGMP in Forwarding**
- ✓ **Variables and Timers**
- ✓ **Encapsulation**
- ✓ **Compatibility with other Versions**

Figure 12.6 *Position of IGMP in the network layer*



Note

***IGMP is a group management protocol.
It helps a multicast router create and
update a list of loyal members related
to each router interface.***

Figure 12.7 *IGMP messages*

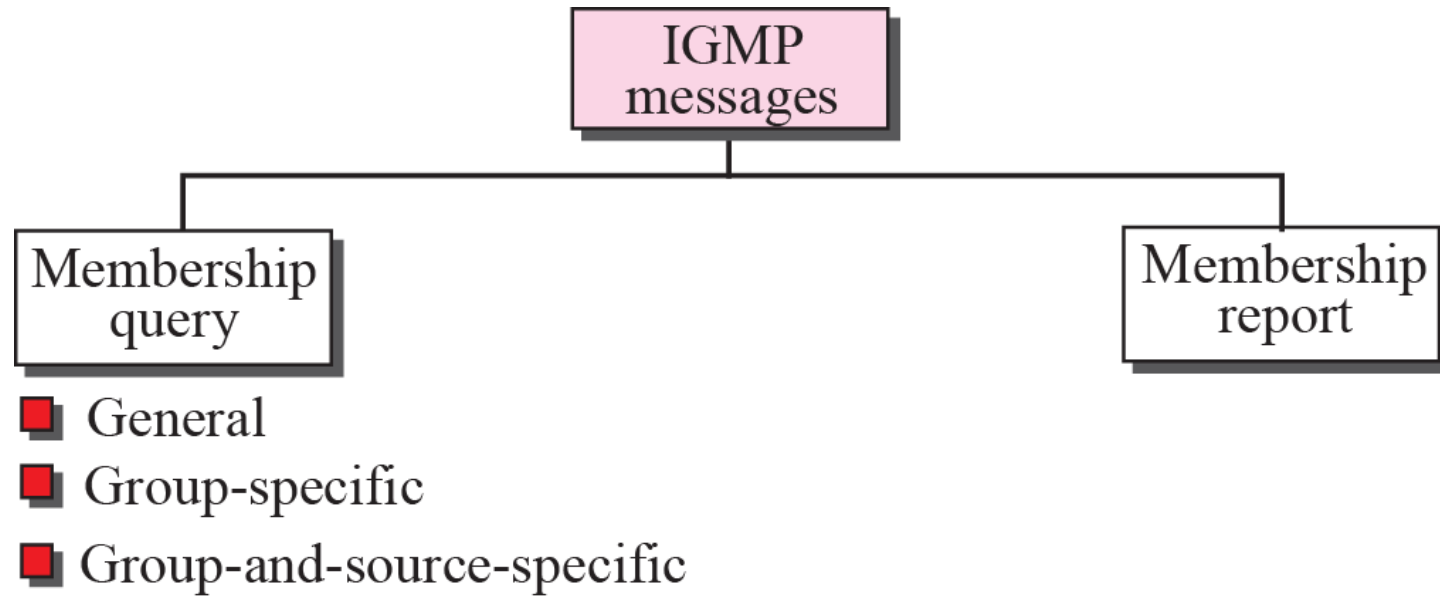


Figure 12.8 *Membership query message format*

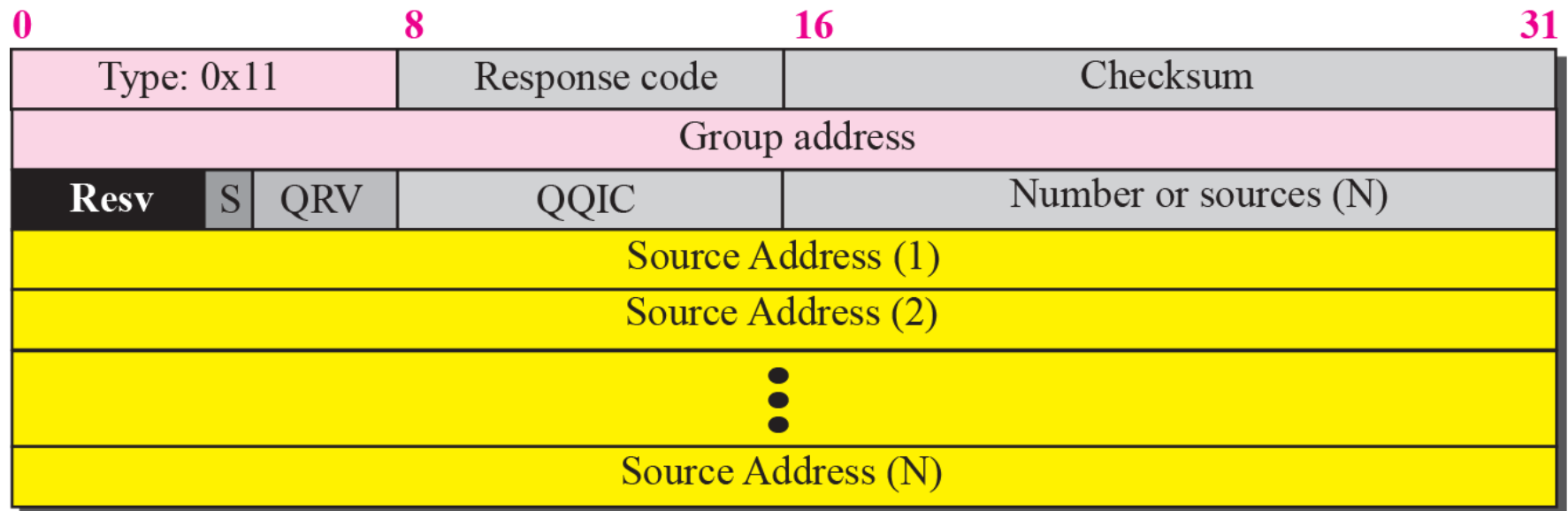
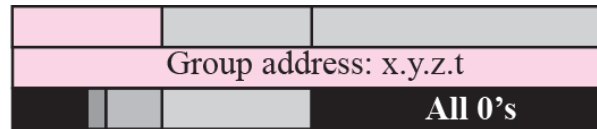


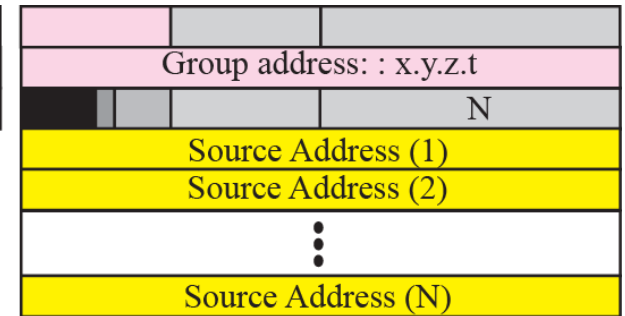
Figure 12.9 *Three forms of query messages*



a. General

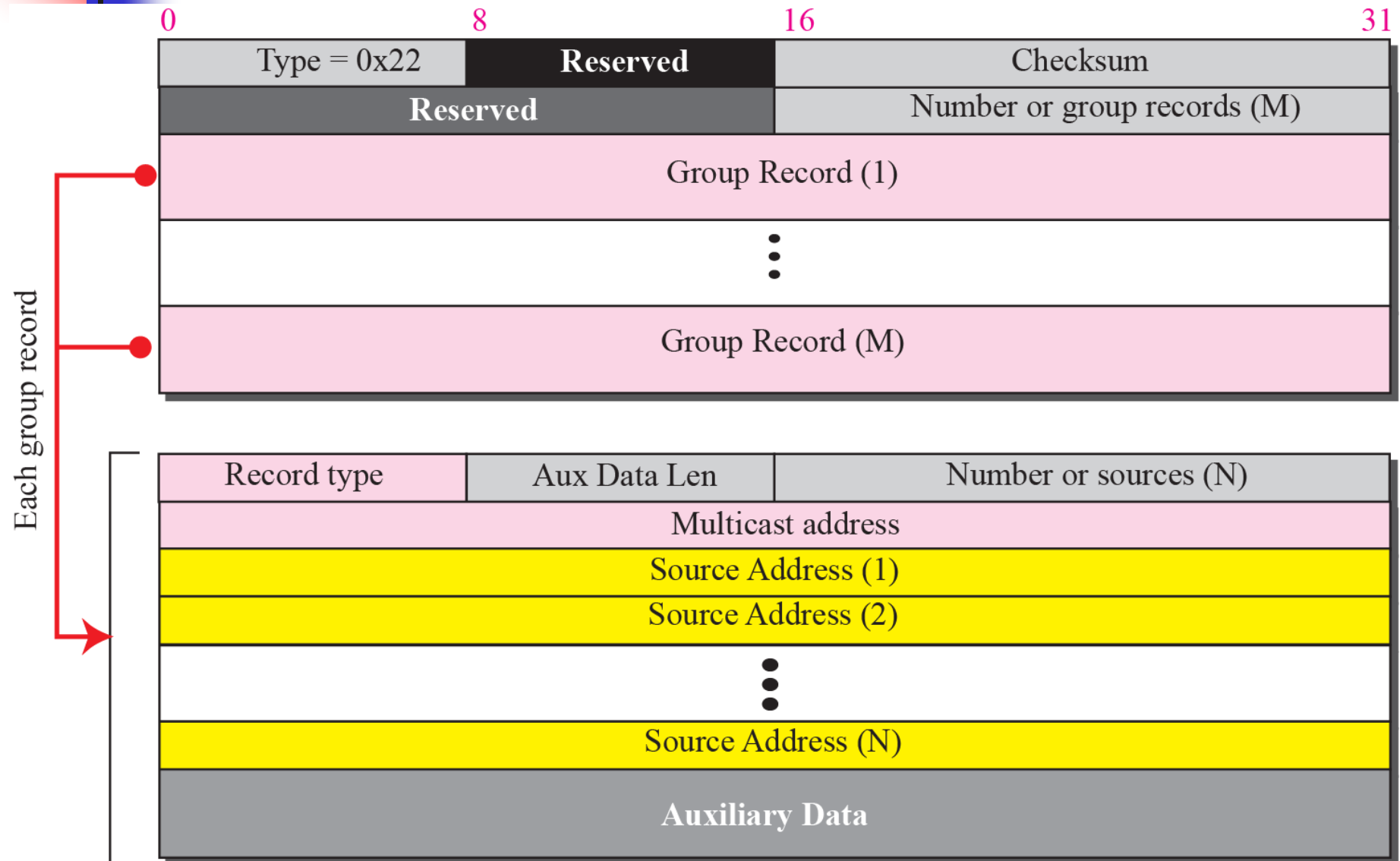


b. Group-specific



c. Group-and-source-specific

Figure 12.10 *Membership report message format*



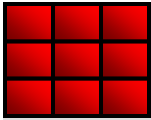


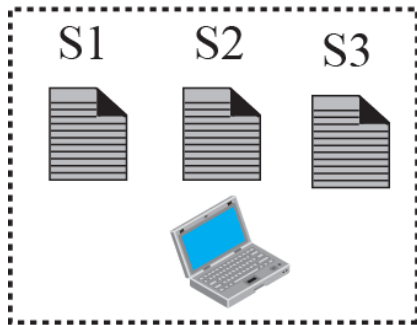
Table 12.3 *Record Type*

<i>Category</i>	<i>Type</i>	<i>Type Value</i>
Current-State-Record	Mode_Is_Include	1
	Mode_Is_Exclude	2
Filter-Mode-Change-Record	Change_To_Include_Mode	3
	Change_To_Exclude_Mode	4
Source-List-Change-Record	Allow_New_Sources	5
	Block_Old_Sources	6

Example 12.4

Figure 12.11 shows a host with three processes: S1, S2, and S3. The first process has only one record; the second and the third processes each have two records. We have used lowercase alphabet to show the source address.

Figure 12.11 *Socket state*



Legend

S: Socket
a, b, ...: Source addresses

States Table

Socket	Multicast group	Filter	Source addresses
S1	226.14.5.2	Include	a, b, d, e
S2	226.14.5.2	Exclude	a, b, c
S2	228.24.21.4	Include	b, c, f
S3	226.14.5.2	Exclude	b, c, g
S3	228.24.21.4	Include	d, e, f

Note

Each time there is a change in any socket record, the interface state will change using the above-mentioned rules.

Example 12.5

We use the two rules described above to create the interface state for the host in Example 12.4. First we found the list of source address for each multicast group.

a. Multicast group 226.14.5.2 has two exclude lists and one include list. The result is an exclude list as calculated below.

$$\text{exclude source list} = \{a, b, c\} \cdot \{b, c, g\} - \{a, b, d, e\} = \{c\}$$

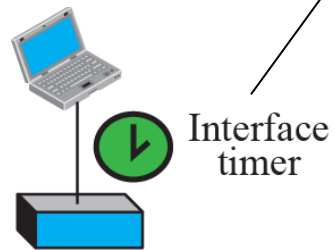
b. Multicast group: 228.24.21.4 has two include lists. The result is an include list as calculated below. We use the plus sign for the union operation.

$$\text{include source list} = \{b, c, f\} + \{d, e, f\} = \{b, c, d, e, f\}$$

Figure 12.12 shows the interface state.

Figure 12.12 *Interface state*

For delaying response to queries (Report Suppression)



Interface state

Multicast group	Group timer	Filter	Source addresses
226.14.5.2		Exclude	c
228.24.21.4		Include	b, c, d, e, f

Figure 12.13 *Sending change state reports*

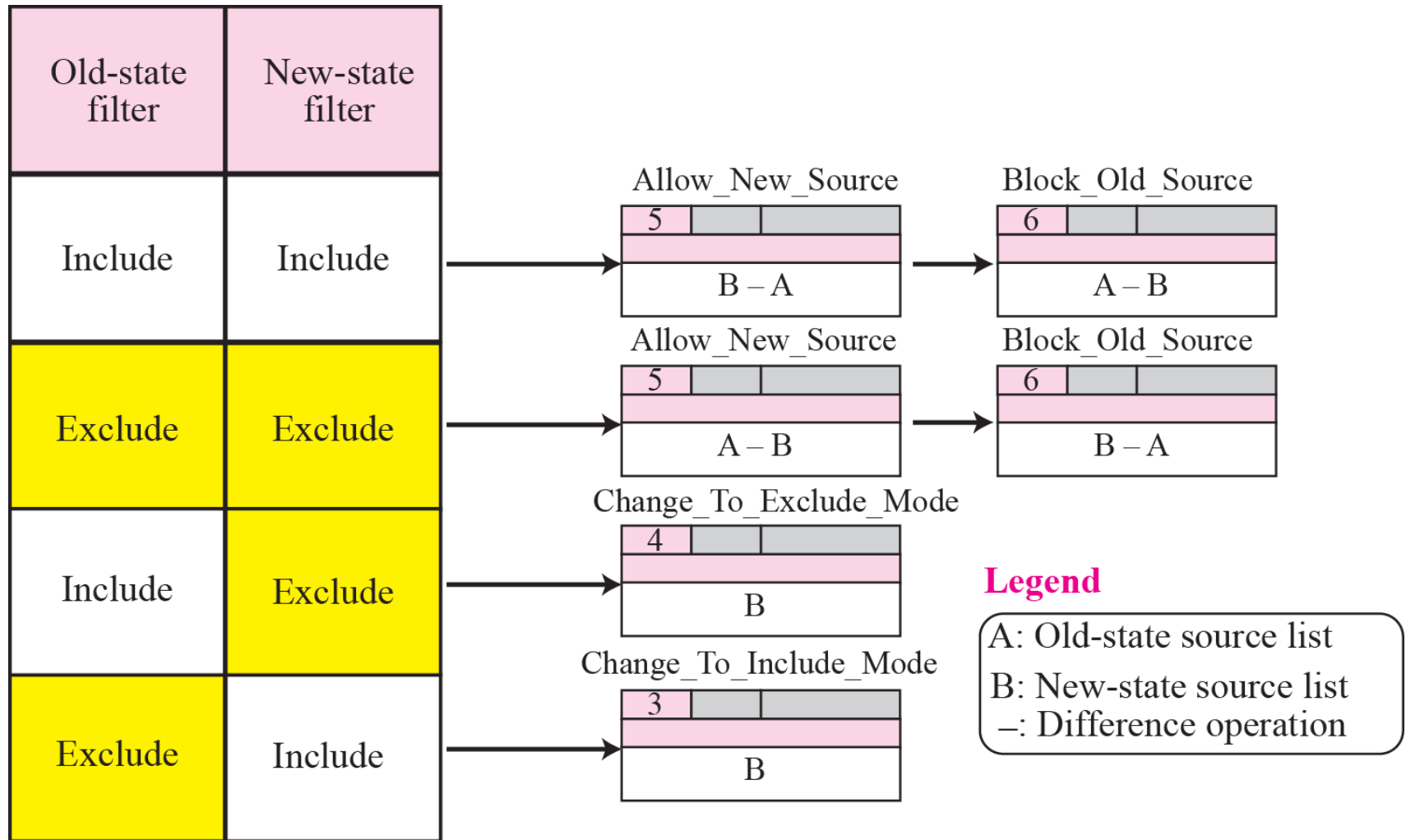
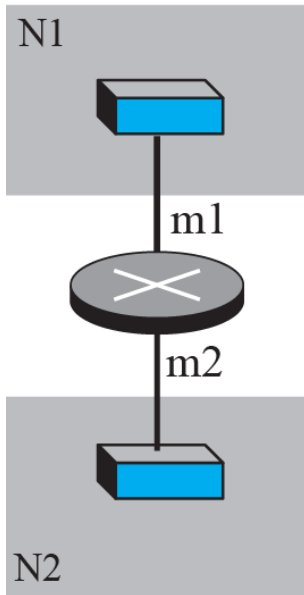


Figure 12.14 Router States



State for interface m1

Multicast group	Timer	Filter	Source addresses
227.12.15.21		Exclude	(a ,) (c ,)
228.21.25.41		Include	(b ,) (d ,) (e ,)

State for interface m2

Multicast group	Timer	Filter	Source addresses
226.10.11.8		Exclude	(b ,)
227.21.25.41		Include	(a ,) (b ,) (c ,)
228.32.12.40		Include	(d ,) (e ,) (f ,)

For sending corresponding queries

Figure 12.15 *Change of state related to general query report*

Legend

{X}: Record with source list X

{X, Y}: Record with source list X with nonzero timer and list Y with zero timer

Include {X}: Record with include filter and sources X

Exclude {X}: Record with exclude filter and source X

Type N {X}: Arrival of report with record of type n and source list X

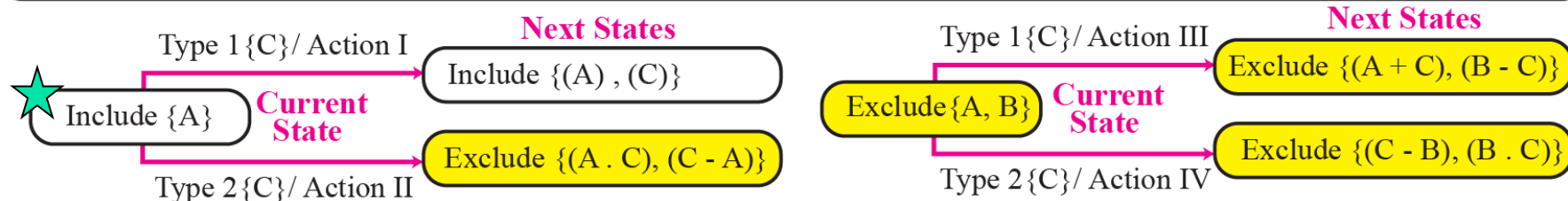
Plus sign: Union operation

Reset: Reset group timer

Reset {X}: Reset timers for X

Dot sign: Intersection operation

Minus sign : Difference operation



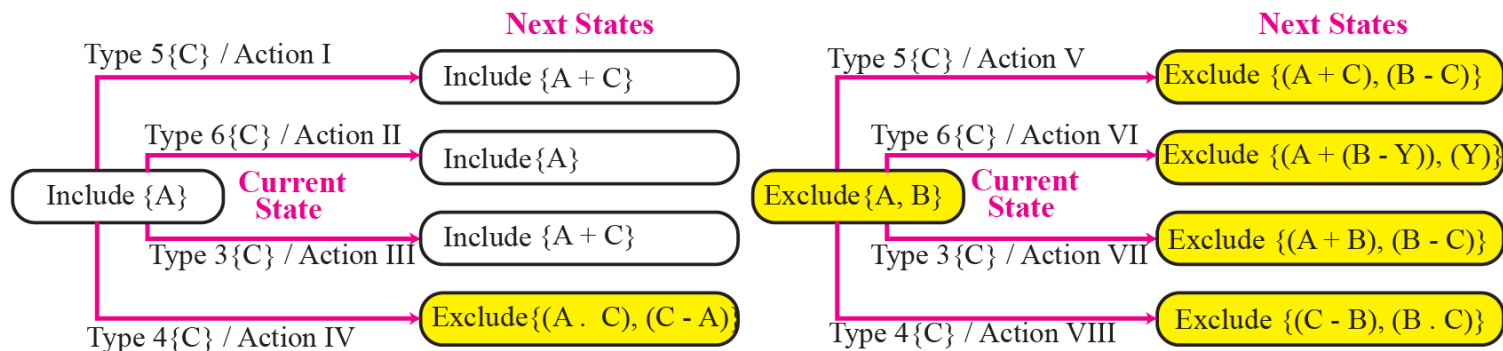
Actions

	Group timer	Sources timers	Delete
I		Reset {C}	{C - A}
II	Reset	Reset {C - A}	
III		Reset {C}	
IV	Reset	Reset {C - A - B}	{A - C} & {B - C}

Figure 12.16 *Change of state related to other report types*

Legend

$\{X, Y\}$: Record with sources X with nonzero timer and sources Y with zero timer
 Include $\{X\}$: Record with include filter and sources X
 Exclude $\{X\}$: Record with exclude filter and source X
 Type N $\{X\}$: Arrival of report with record of type N and sources X
 Q-G-S $\{X\}$: Send group-and-source-specific query with sources X
 Dot ($\cdot$): Intersection operation
 Plus ($+$): Union operation
 $\{X\}$: Record with sources X
 Q-G: Send group-specific query
 Reset: Reset group timer
 Reset $\{X\}$: Reset timers for source X
 Minus ($-$): Difference operation



Actions

	Group timer	Sources timers	Delete	Send G-S query	Send G-S-S query
I		Reset {C}			
II					Q-G-S {A · C}
III		Reset {C}			Q-G-S {A - C}
IV	Reset		{A - C}		Q-G-S {A · C}
V		Reset {C}			
VI		Reset {C - A - B}			Q-G-S {C - B}
VII		Reset {C}		Q-G	Q-G-S {A - C}
VIII	Reset		{A - C} & {B - C}		Q-G-S {A - B}

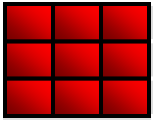
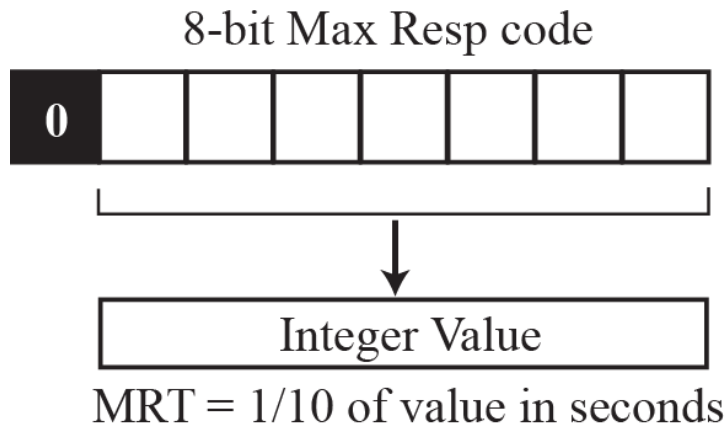


Table 12.4 *Forwarding Recommendation of IGMPv3*

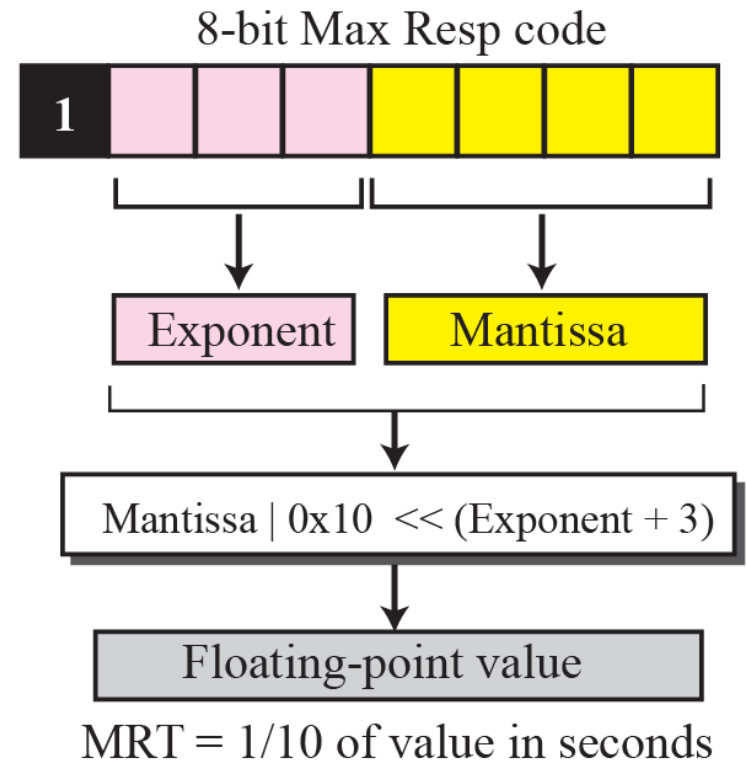
<i>Filter Mode</i>	<i>Source Address</i>	<i>Source Timer Value</i>	<i>Recommendation</i>
Include	In the list	greater than zero	Forward
Include	In the list	zero	Do not forward
Include	Not in the list		Do not forward
Exclude	In the list	greater than zero	Forward
Exclude	In the list	zero	Do not forward
Exclude	Not in the list		Forward

Figure 12.17 *Calculation of maximum response time*



Legend

MRT: maximum response time
| : bit-wise OR operation
<< : bit-wise shift-left operation



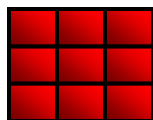


Table 12.5 *Destination IP Addresses*

<i>Message Type</i>	<i>IP Address</i>
General Query	224.0.0.1
Other Queries	Group address
Report	224.0.0.22

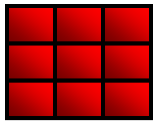


Table 12.6 *Messages in Versions 1 and 2*

<i>Version</i>	<i>Type Value</i>	<i>Message Type</i>
1	0x11	Query
	0x12	Membership Report
2	0x11	Query
	0x16	Membership Report
	0x17	Leave Group

12-4 MULTICAST ROUTING

Now we show how information collected by IGMP is disseminated to other routers using multicast routing protocols. However, we first discuss the idea of optimal routing, common in all multicast protocols. We then give an overview of multicast routing protocols.

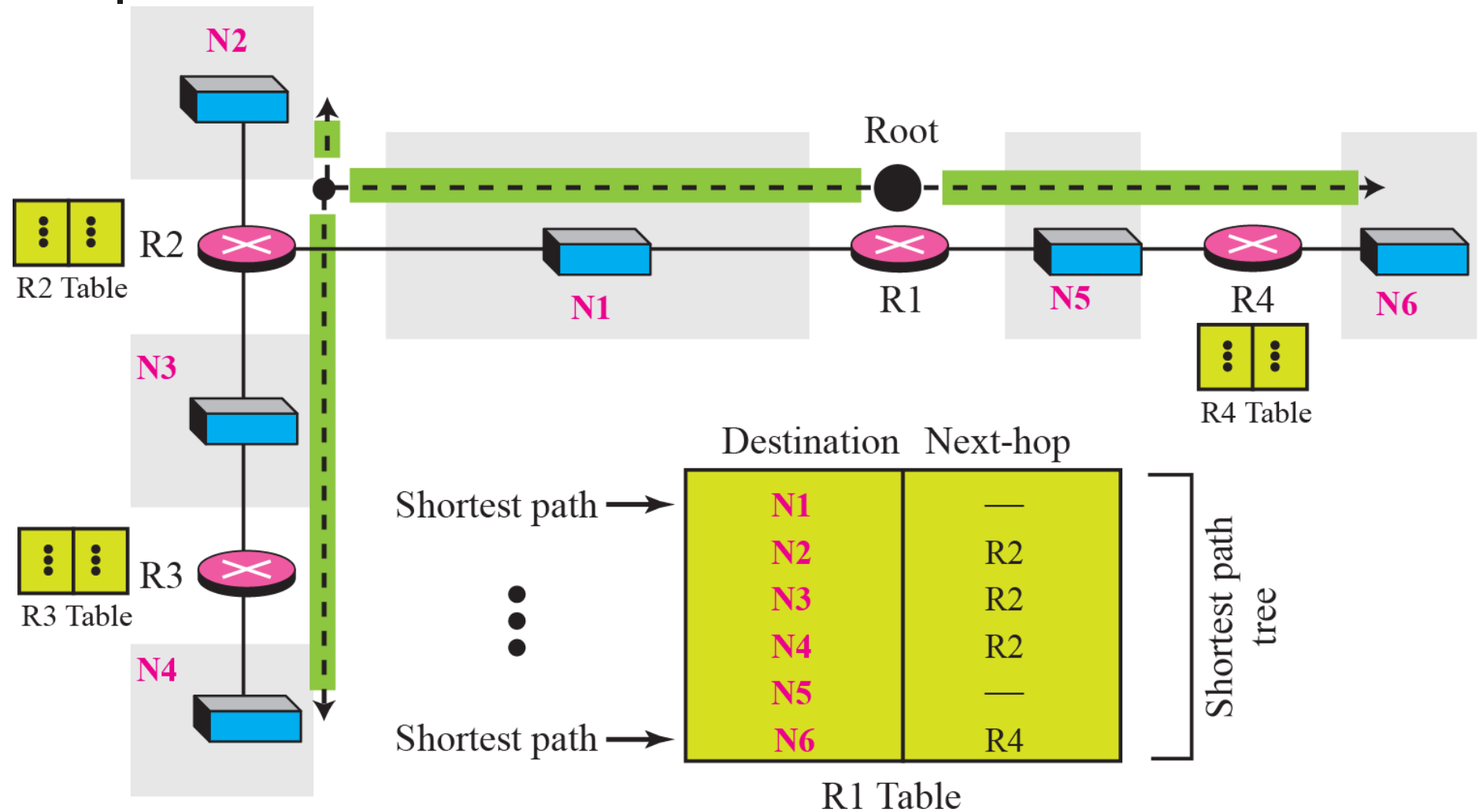
Topics Discussed in the Section

- ✓ **Optimal Routing: Shortest Path Trees**

Note

In unicast routing, each router in the domain has a table that defines a shortest path tree to possible destinations.

Figure 12.18 *Shortest path tree in multicast routing*



Note

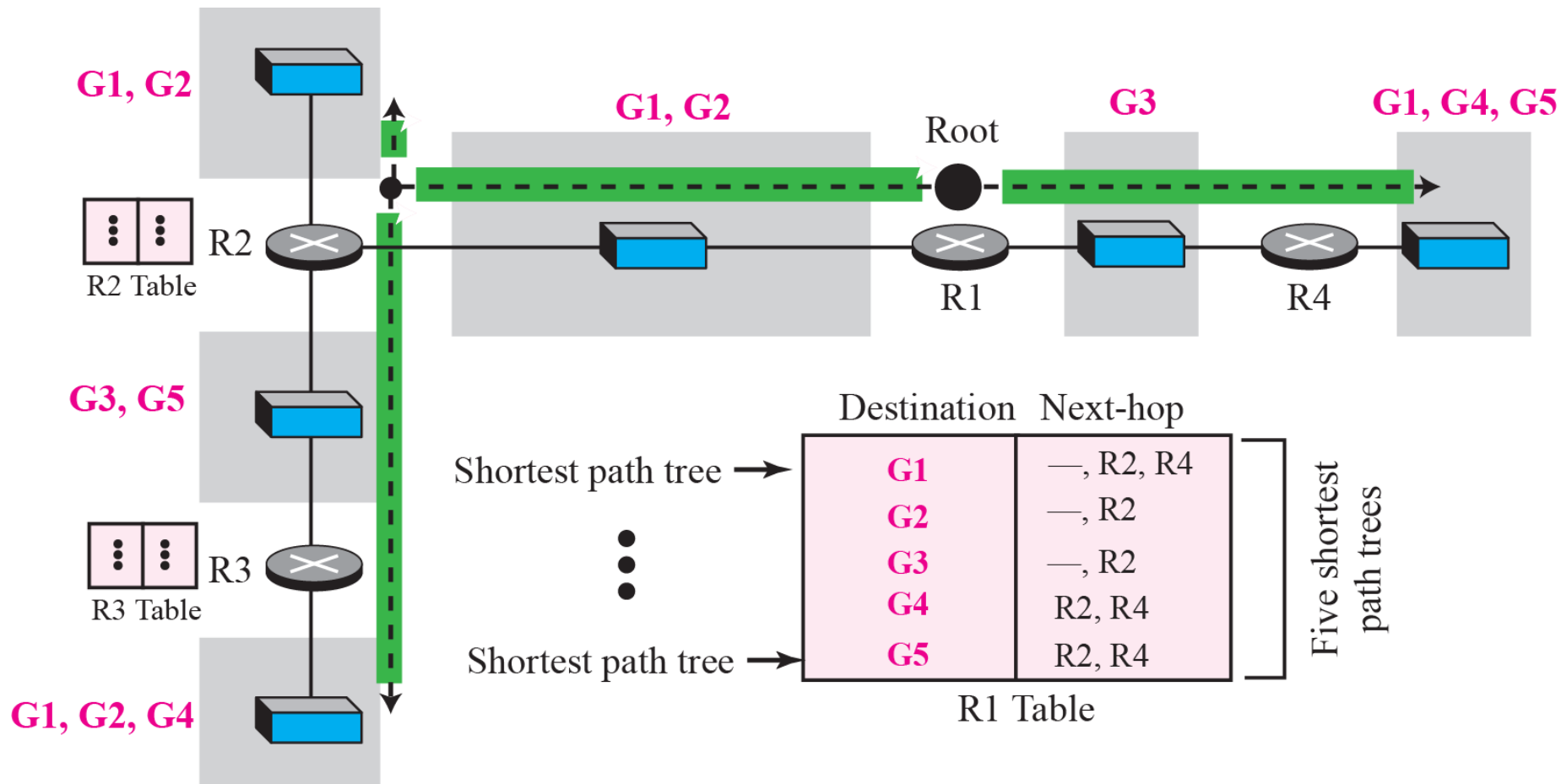
In multicast routing, each involved router needs to construct a shortest path tree for each group.

Note

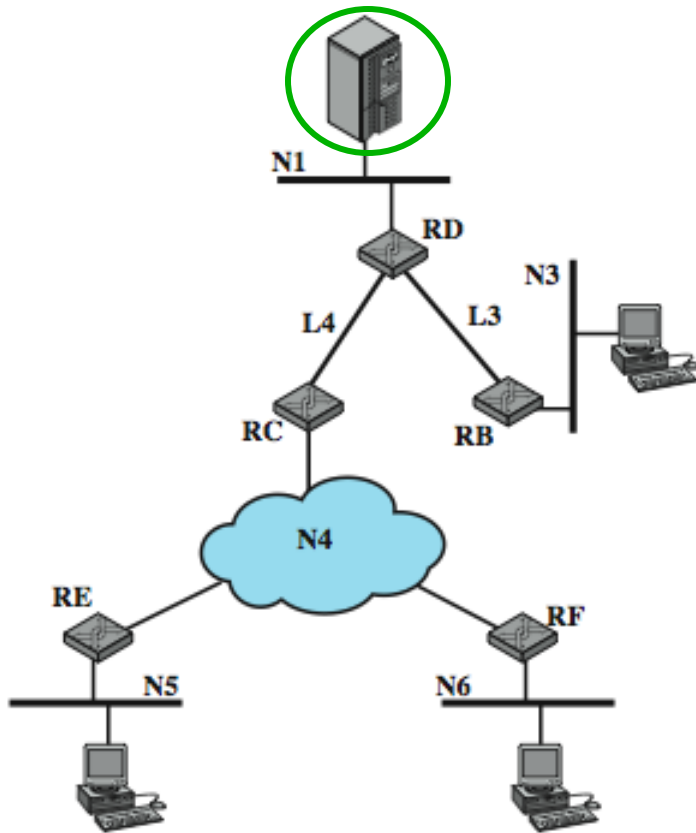
*In the source-based tree approach, each router needs to have one shortest path tree for each group **and source**.*

Figure 12.19 *Source-based tree approach*

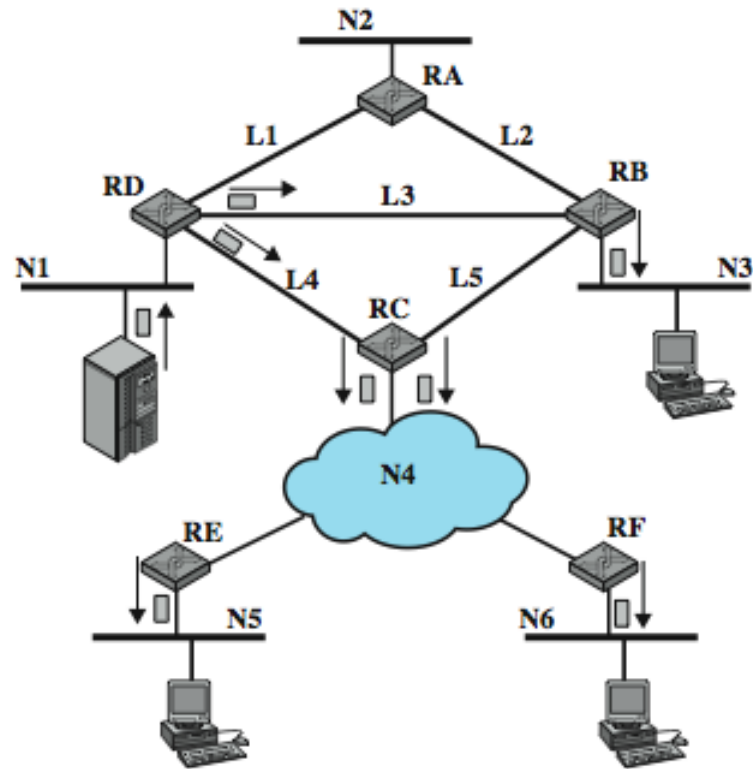
注意：課本中此段的文字敘述有問題，請參考網頁的參考資料。



E.g. Source-based Tree (1)

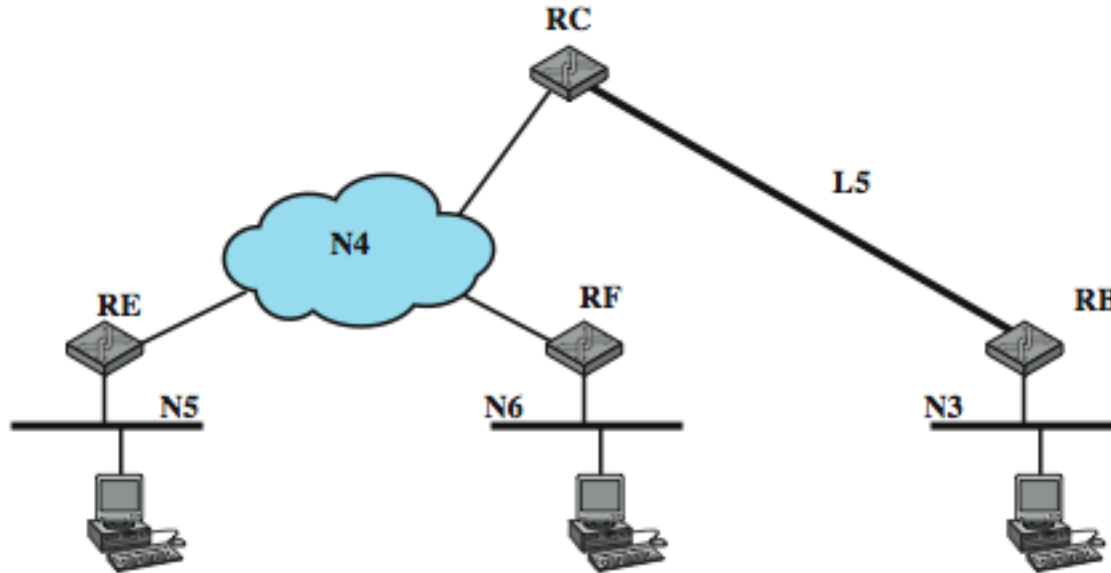


(a) Spanning tree from source to multicast group



(b) Packets generated for multicast transmission

E.g. Source-based Tree (2)

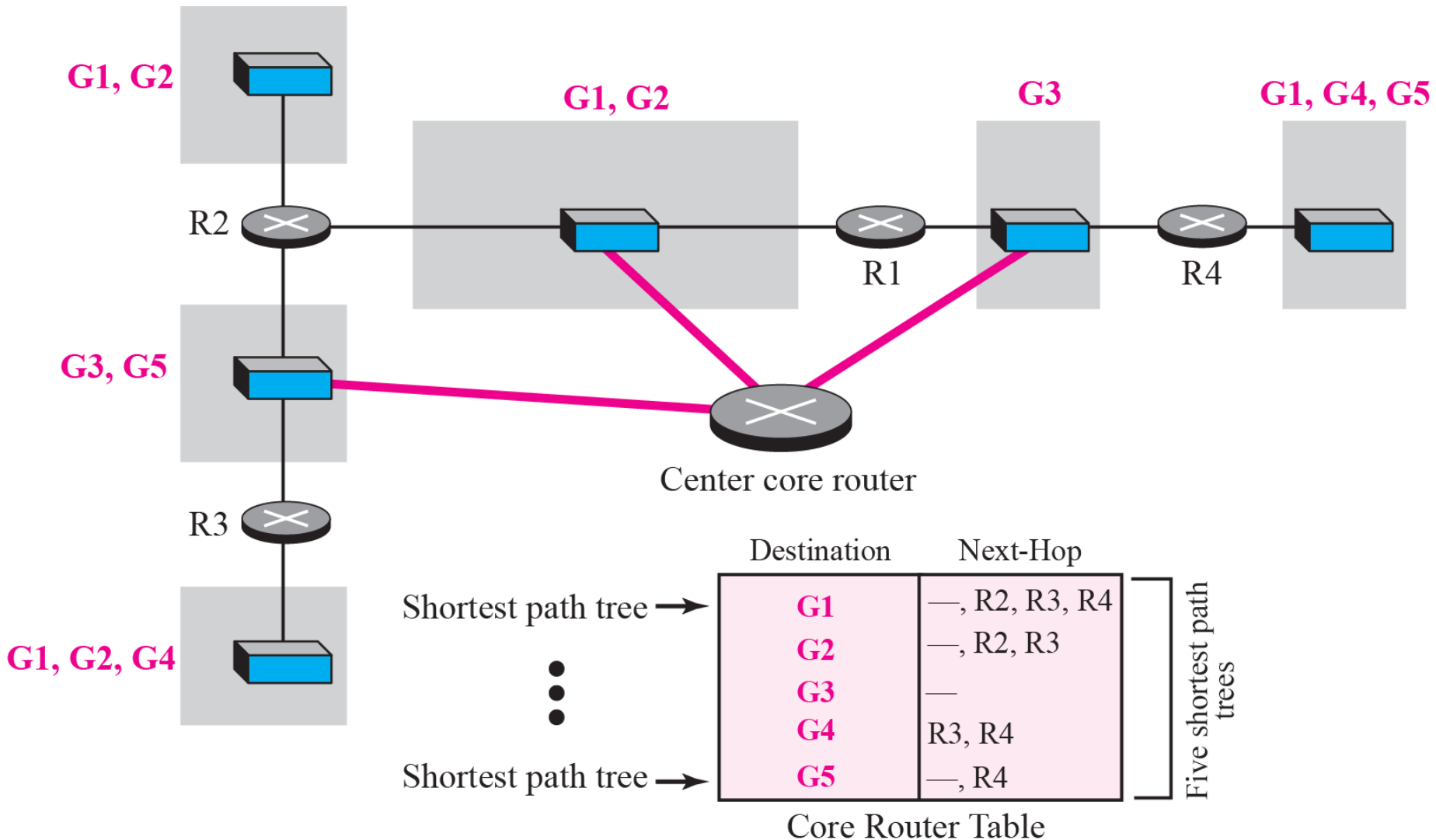


Spanning Tree from Router C to Multicast Group

Note

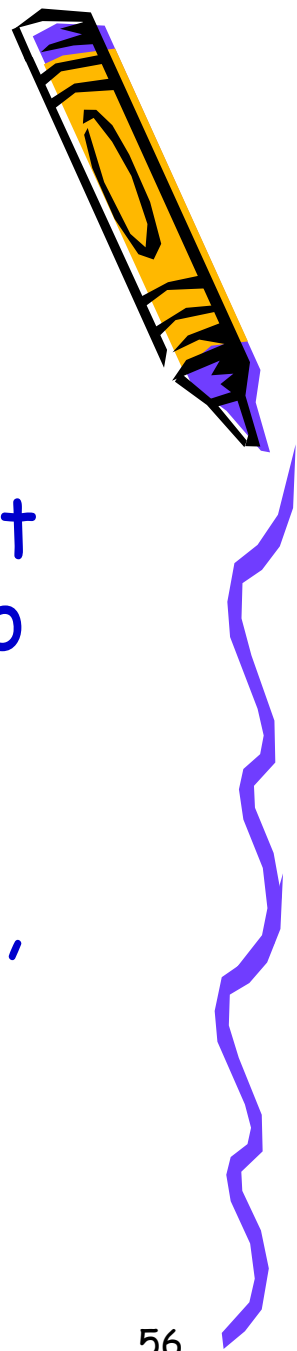
In the group-shared tree approach, only the core router, which has a shortest path tree for each group, is involved in multicasting.

Figure 12.20 *Group-shared tree approach*



Group-Shared Tree

- If a router receives a multicast packet, it encapsulates the packet in a unicast packet and sends it to the **core router**
- The core router removes the multicast packet from its capsule, and consults its routing table to route the packet



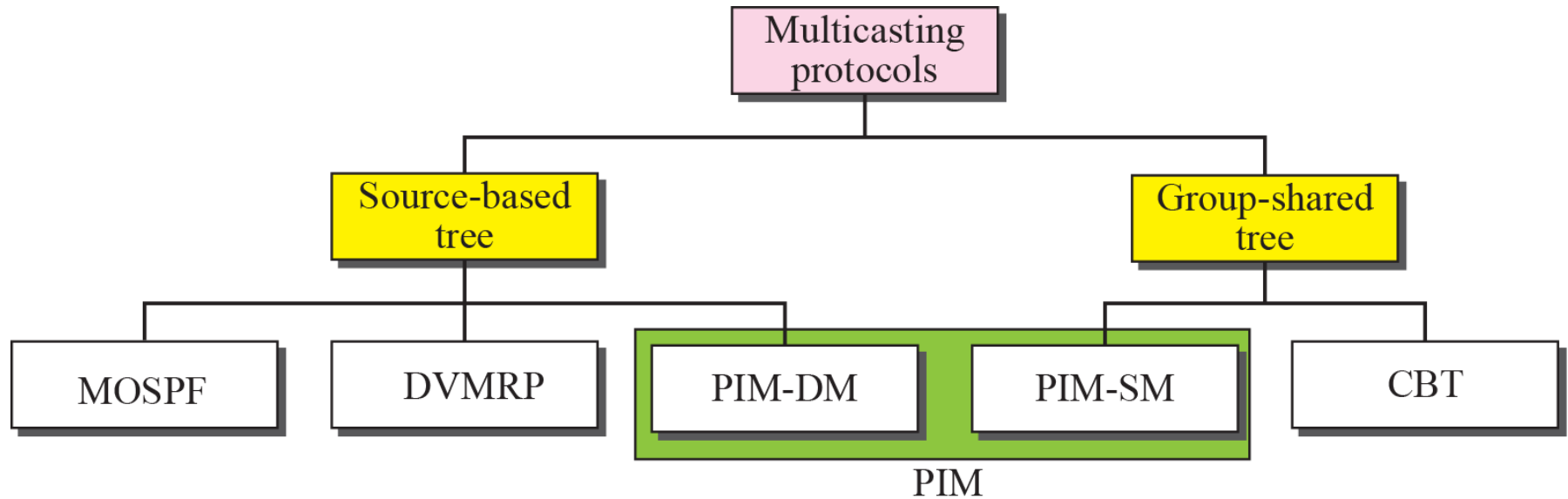
12-4 ROUTING PROTOCOLS

During the last few decades, several multicast routing protocols have emerged. Some of these protocols are extensions of unicast routing protocols; some are totally new. We discuss these protocols in the remainder of this chapter. Figure 12.21 shows the taxonomy of these protocols.

Topics Discussed in the Section

- ✓ **Multicast Link State Routing: MOSPF**
- ✓ **Multicast Distance Vector: DVMRP**
- ✓ **Core-Based Tree: CBT**
- ✓ **Protocol Independent Multicast: PIM**

Figure 12.21 *Taxonomy of common multicast protocols*





Note

Multicast link state routing uses the source-based tree approach.

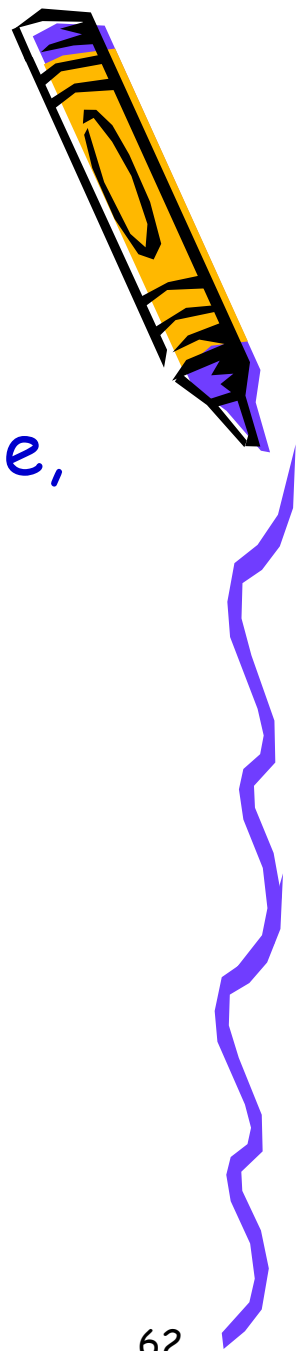
MOSPF (1)

- Group membership LSA is flooded throughout the AS
- The router calculates the shortest path trees on demand (when it receives the first multicast packet)
- MOSPF is a data-driven protocol; the first time an MOSPF router see a datagram with a given source and group address, the router constructs the Dijkstra shortest path tree



MOSPF (2)

- The shortest path tree is made all at once instead of gradually (i.e. pre-made, pre-pruned, ready to use)



MOSPF with Areas (1)

- **Group management**
 - Group-membership LSA is flooded in the same area.
 - **Inter-area multicast forwarders** (area border routers) summarize their attached areas' group membership to the backbone.
- **Data routing**
 - Introduction of the **wild-card multicast receivers** (area border routers)



MOSPF with Areas (2)

- Data routing (cont)
 - In the presence of OSPF areas, during tree pruning care must be taken so that the branches leading to other areas remain, since it is unknown whether there are group members in these (remote) areas.

For this reason, only those branches having no group members nor **wild-card multicast receivers** are pruned when producing the datagram shortest-path tree.



MOSPF with Areas (3)

- Data routing (cont)
 1. Source area: building **intra-area shortest path tree** (forward cost) with leaf nodes including wild-card multicast receivers.
 2. Backbone area: each wild-card multicast receiver of the source area calculates the shortest path from the source to the multicast forwarders (with group members) of other areas using **the reverse cost**.

(You must know the reason why using **the reverse cost** in this case)



MOSPF with Areas (4)

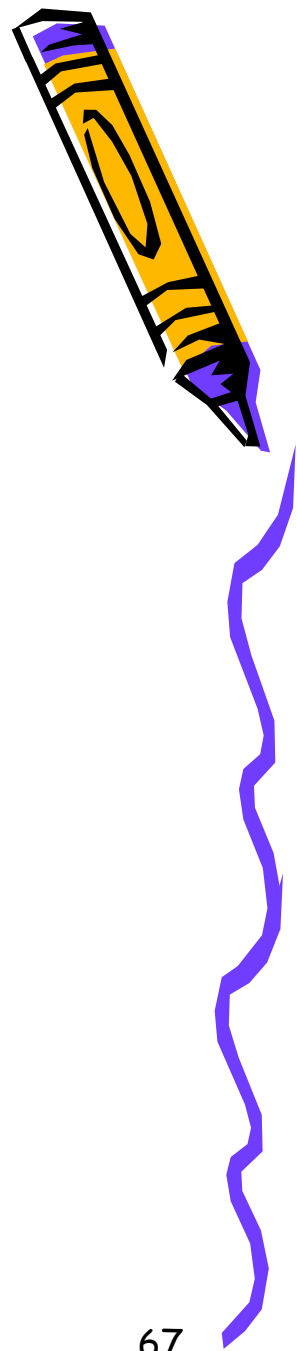
- Data routing (cont)

E.g. In Figure 4 of the sample MOSPF area configuration in the supplementary document, RT3 and RT4 can calculate and compare to determine which one of them should construct the shortest path from the source to RT7, RT10, and RT11 (the multicast forwarders in non-source areas). The result tree is shown in Figure 9 in the supplementary document.

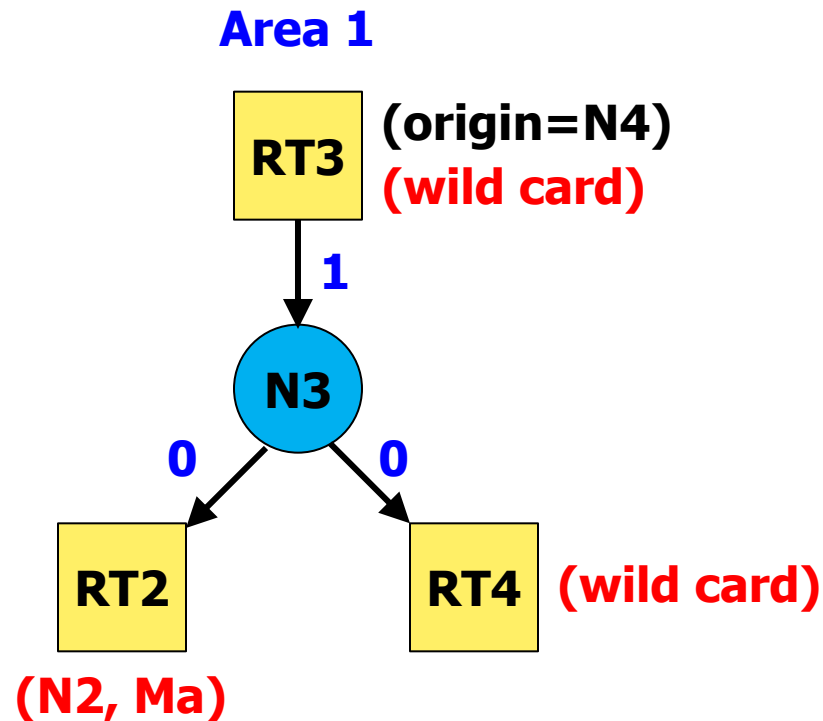


MOSPF with Areas (5)

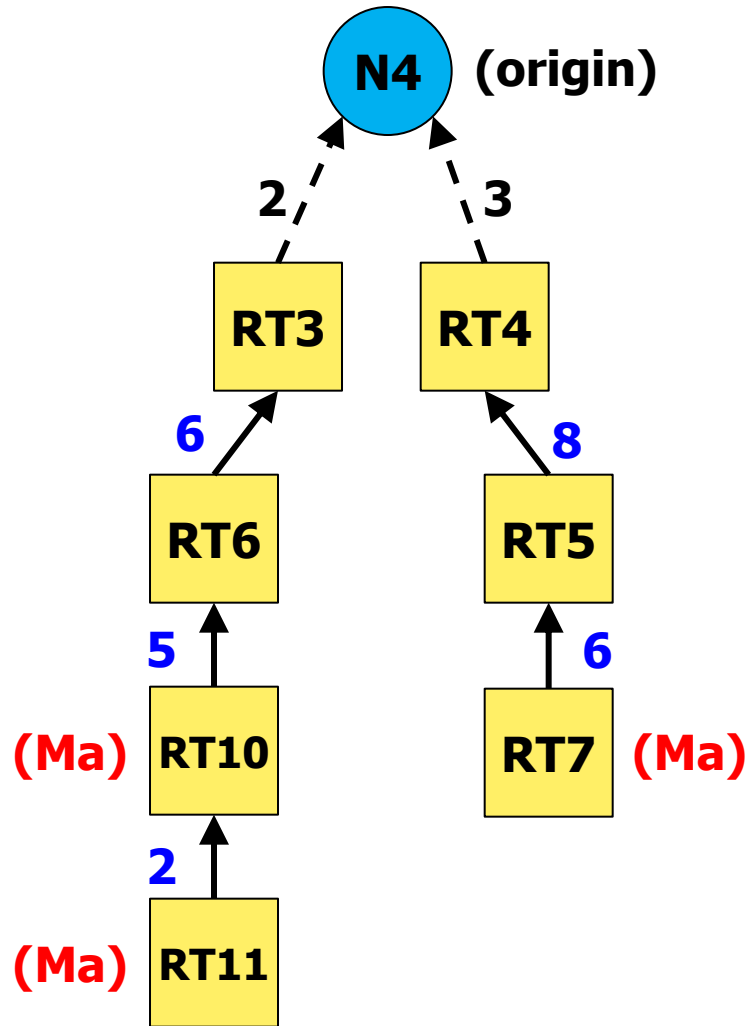
- Data routing (cont)
 3. Destination areas: The corresponding multicast forwarder (e.g. RT7 in area 2) constructs the shortest path using the reverse cost to each network (with group members) in its own area.



MOSPF Tree in Source Area

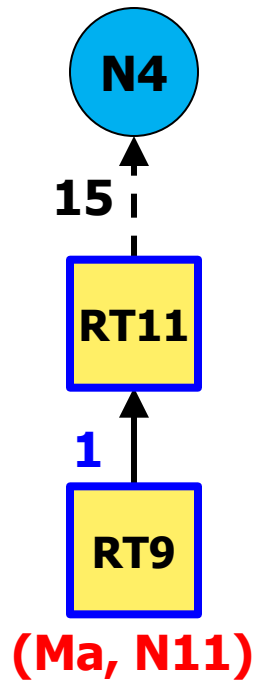


MOSPF Tree in Backbone Area

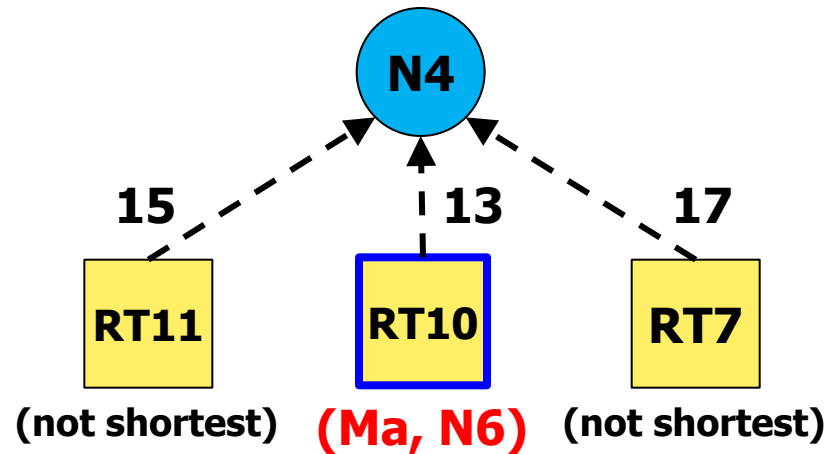


MOSPF Tree in Destination Area

Area 3
(origin)



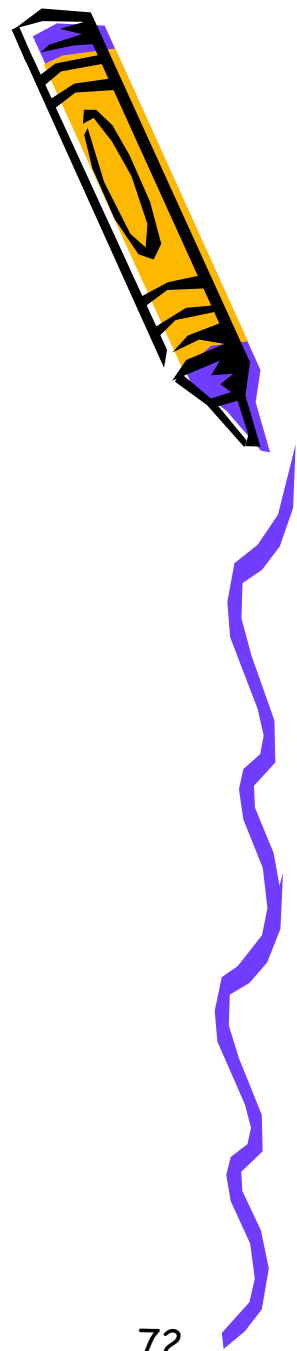
Area 2
(origin)



Multicast Distance Vector Routing

- 4 decision-making strategies

1. Flooding
2. Reverse Path Forwarding (RPF)
3. Reverse Path Broadcasting (RPB)
4. Reverse Path Multicasting (RPM)





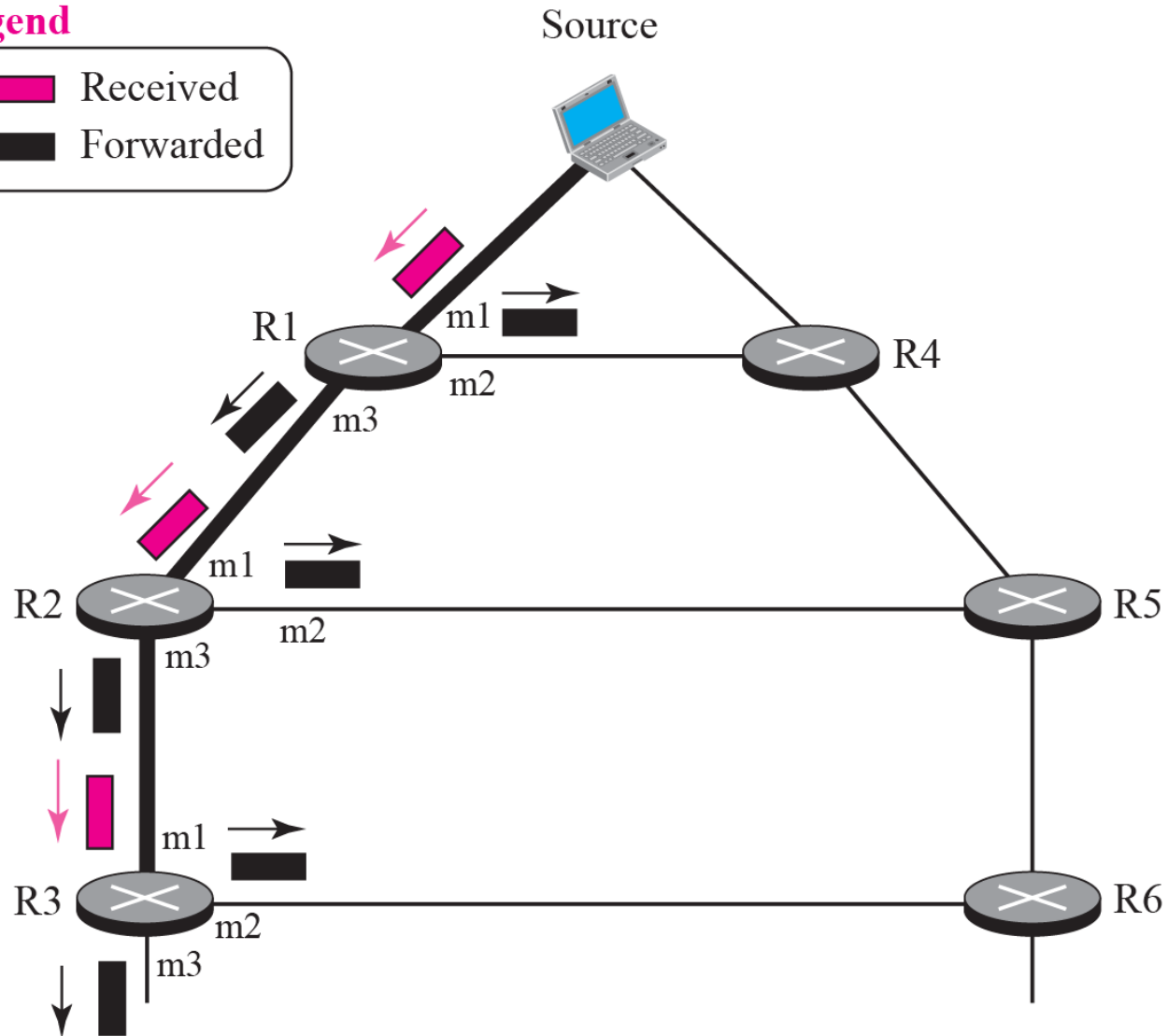
Note

***Flooding broadcasts packets, but
creates loops in the systems.***

Figure 12.22 *RPF*

Legend

- Received
- Forwarded



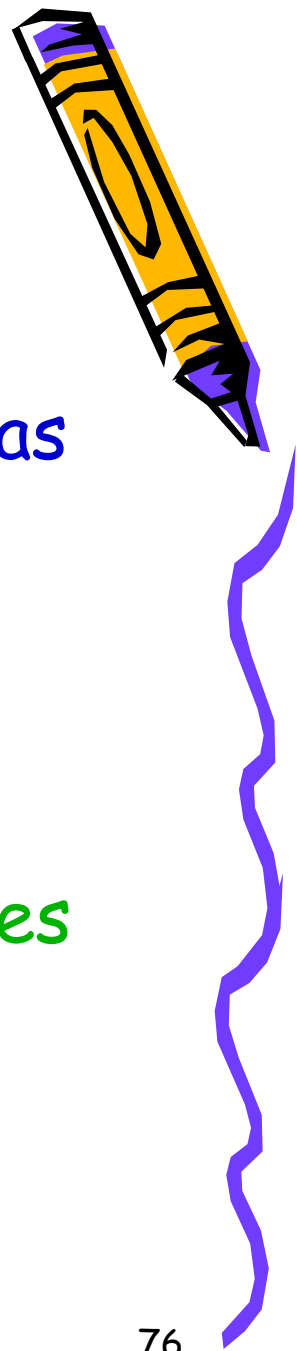
Reverse Path Forwarding (1)

- To prevent loops, only one copy is forwarded; the other copies are dropped.
- In RPF, a router forwards only the copy that has traveled the shortest path from the source to the router.
- The router extracts the source address of the multicast packet and consults its unicast routing table.



Reverse Path Forwarding (2)

- If the packet has just come from the hop defined in the table, the packet has traveled the shortest path from the source to the router because the shortest path is reciprocal in unicast distance vector routing protocols.
- If a packet leaves the router and comes back again, it has not traveled the shortest path.





Note

***RPF eliminates the loop in the
flooding process.***

E.g. No Loops in RPF

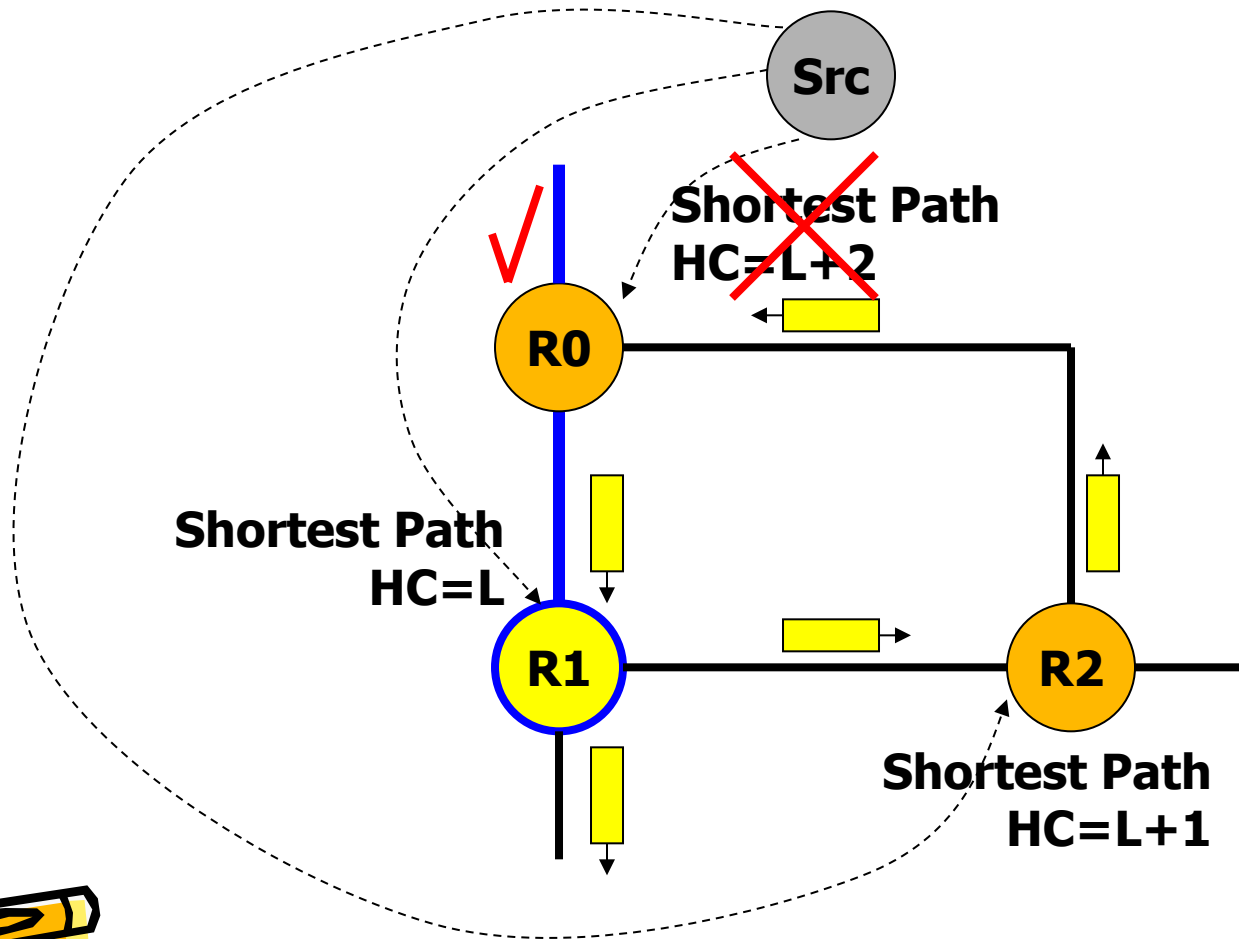


Figure 12.23 *Problem with RPF*

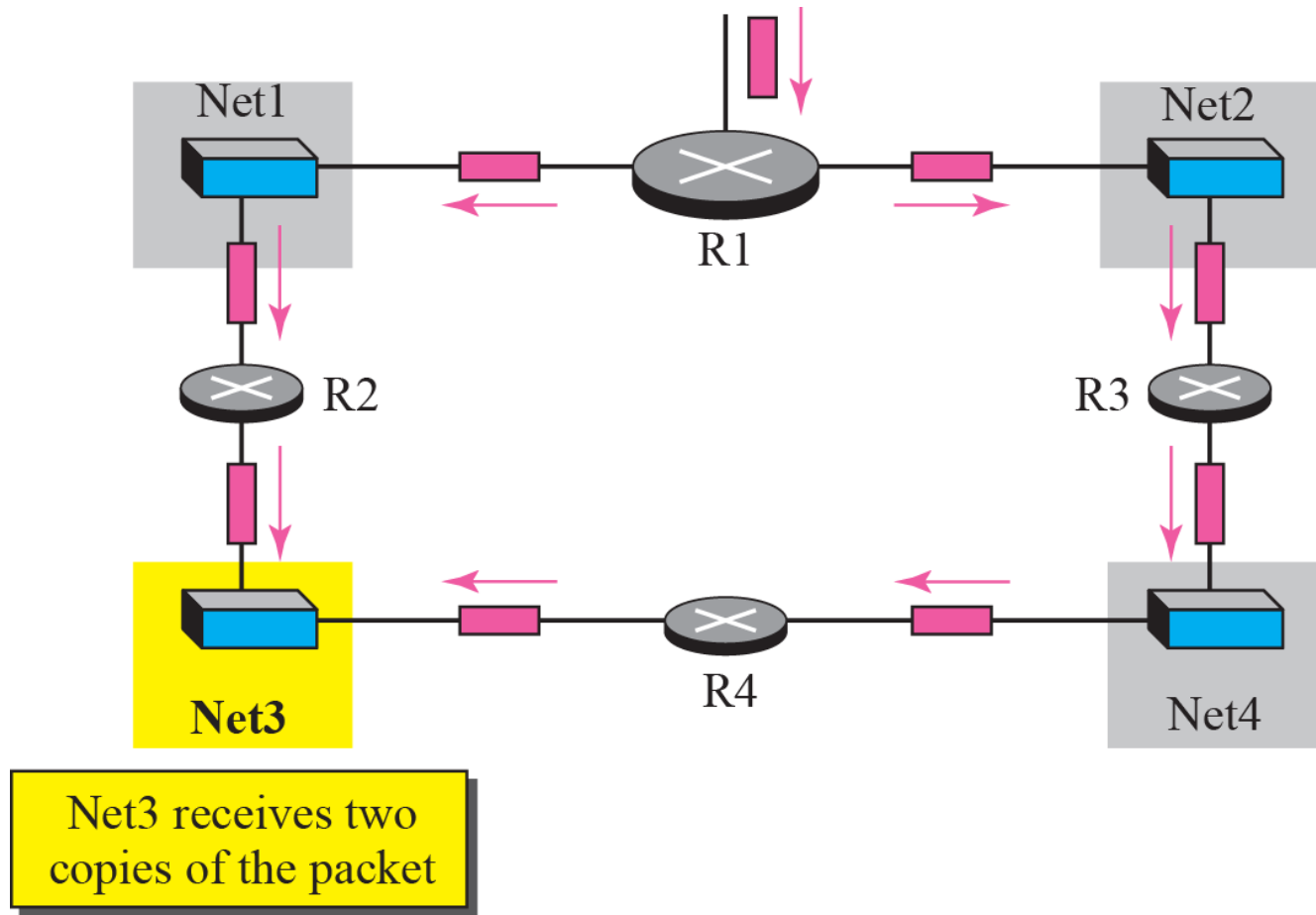
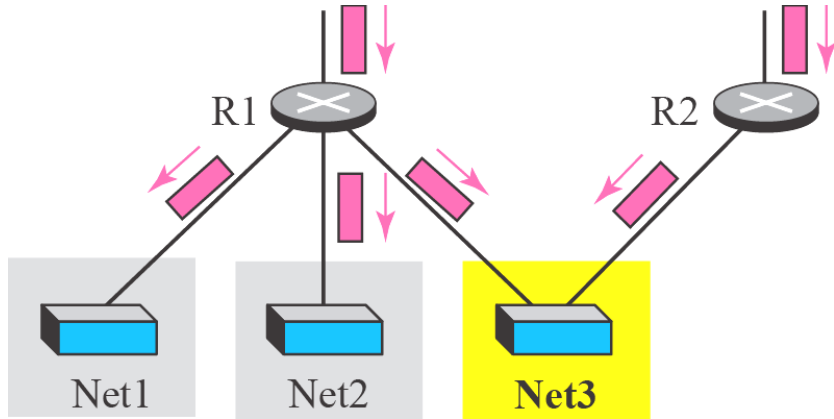
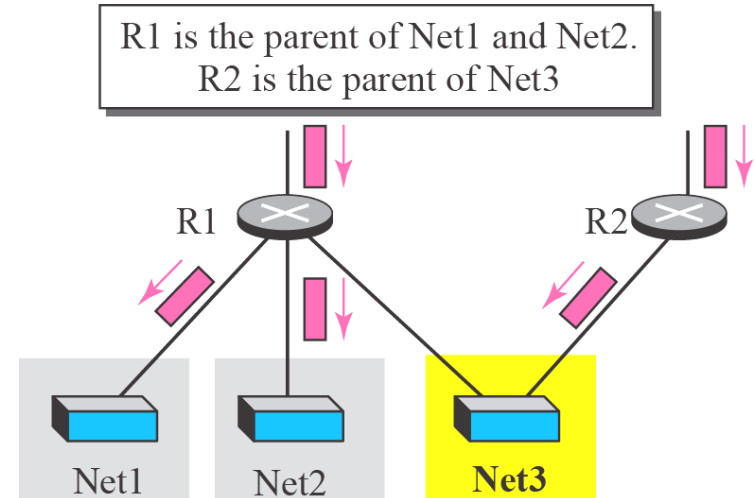


Figure 12.24 *RPF versus RPB*



a. RPF



b. RPB

Reverse Path Broadcasting (1)

- RPF guarantees that each network receives a copy of the multicast packet without formation of loops
- However, RPF does not guarantee that each network receives only one copy
- To eliminate duplication, we must define only one parent router (**designated parent router**) for each network



Reverse Path Broadcasting (2)

- In RPB, for each source, the router sends the packet only out of those interfaces for which it is the designated parent
- The designated parent router can be the router with the shortest path to the source
 - Because routers periodically send updating packets to each other (in RIP), they can easily determine which router in the neighborhood has the shortest path to the source.

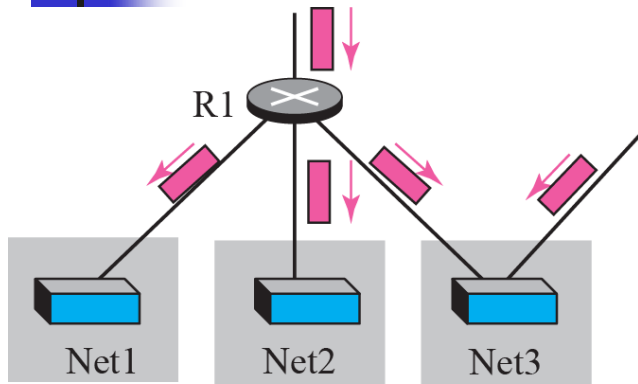


Note

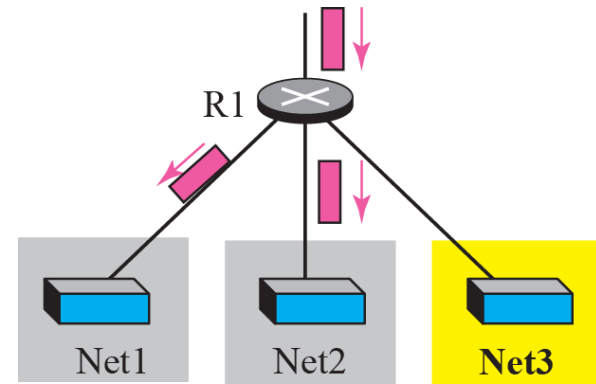
RPB creates a shortest path broadcast tree from the source to each destination.

It guarantees that each destination receives one and only one copy of the packet.

Figure 12.25 *RPF, RPB, and RPM*



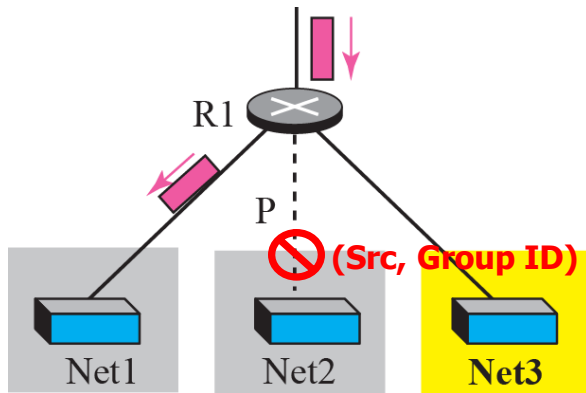
a. RPF



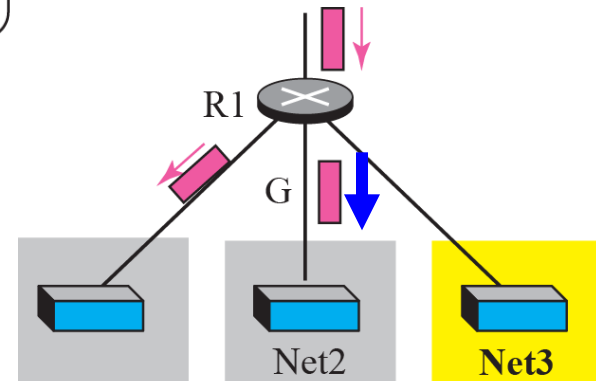
b. RPB

Legend

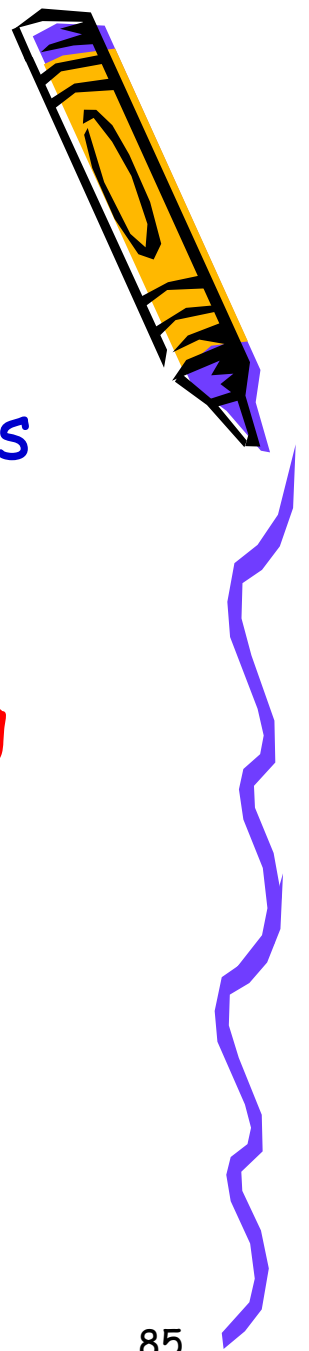
P: Pruned route
G: Grafted route



c. RPM (after pruning)



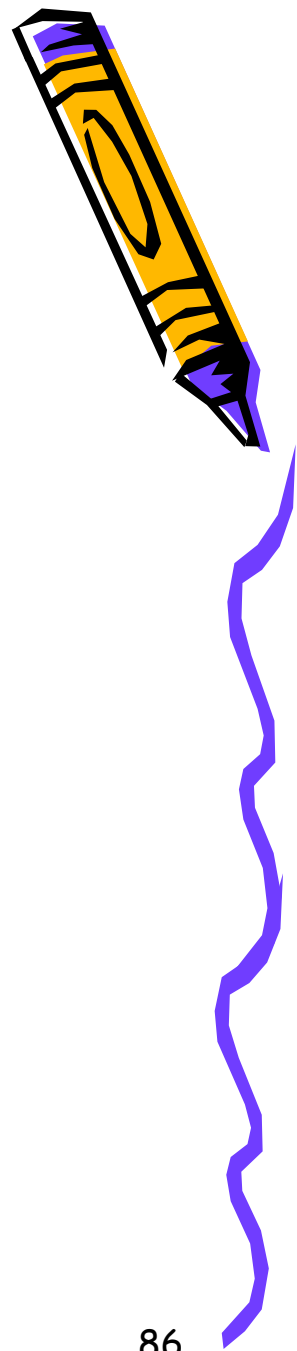
d. RPM (after grafting)



Reverse Path Multicasting (1)

- To increase efficiency, the multicast packet must reach only those networks that have active members for that particular group
- RPM adopts the procedures of **Pruning** and **Grafting**
- **Pruning**
 - The designated parent router of each network is responsible for holding the membership information (through IGMP)





Reverse Path Multicasting (2)

- The router sends a prune message to the upstream router so that it can prune the corresponding interface
- That is, the upstream router can stop sending multicast message for this group through that interface
- **Grafting**
 - The graft message forces the upstream router to resume sending the multicast messages

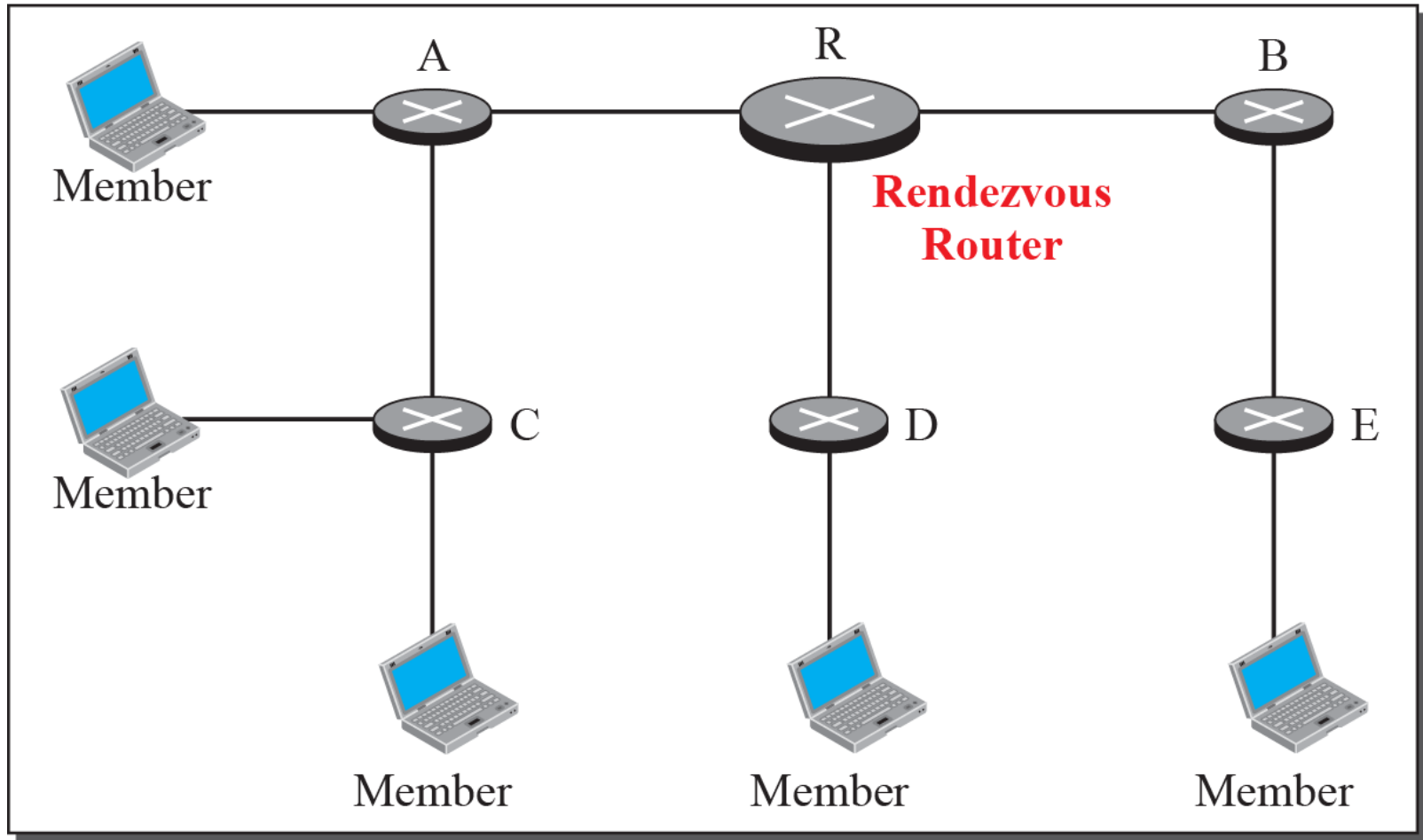


Note

RPM adds pruning and grafting to RPB to create a multicast shortest path tree that supports dynamic membership changes.

Figure 12.26 *Group-shared tree with rendezvous router*

Shared Tree





Formation of CBT tree (1)

- After the **rendezvous point** is selected, every router is informed of the unicast address of the selected router
- Each router sends a unicast join message to show that it wants to join the group
- This message passes through all routers that are located between the sender and the rendezvous router



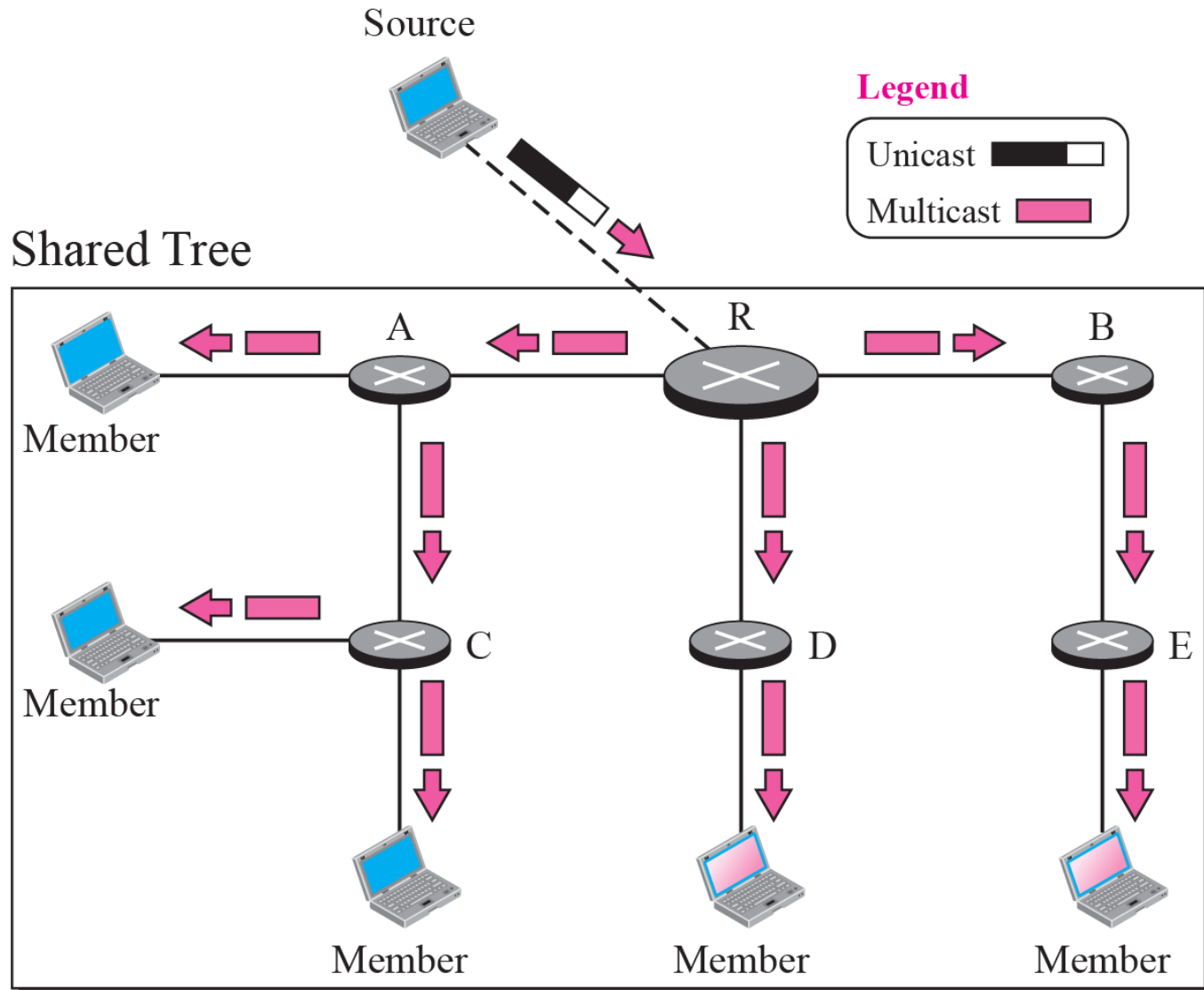


Formation of CBT tree (2)

- Each intermediate router extracts the necessary information from the message
 - Unicast address of the sender
 - Interface through which the packet has arrived
- Every router knows its upstream router and the downstream router
- If a router wants to leave the group, it sends a leave message to its upstream router, ...



Figure 12.27 *Sending a multicast packet to the rendezvous router*



Note

In CBT, the source sends the multicast packet (encapsulated in a unicast packet) to the core router. The core router decapsulates the packet and forwards it to all interested interfaces.

Comparisons

- The tree for **DVMRP** and **MOSPF** is made from the root up
- The tree for **CBT** is formed from the leaves down
- In **DVMRP**, the tree is first made (broadcasting) and then pruned
- In **CBT**, the joining gradually makes the tree, and the source in **CBT** may or may not be part of the tree



Note

PIM-DM is used in a dense multicast environment, such as a LAN.

PIM-DM

- It is used when there is a possibility that each router is involved in multicasting (dense mode)
- In this environment, the use of a protocol that broadcasts the packet is justified because almost all routers are involved in the process



Note

PIM-DM uses RPF and pruning/grafting strategies to handle multicasting.

However, it is independent from the underlying unicast protocol.

Note

PIM-SM is used in a sparse multicast environment such as a WAN.



Note

PIM-SM is similar to CBT but uses a simpler procedure.

12-6 MBONE

Multimedia and real-time communication have increased the need for multicasting in the Internet. However, only a small fraction of Internet routers are multicast routers. Although this problem may be solved in the next few years by adding more and more multicast routers, there is another solution for this problem. The solution is tunneling. The multicast routers are seen as a group of routers on top of unicast routers. The multicast routers may not be connected directly, but they are connected logically.

Figure 12.28 *Logical tunneling*

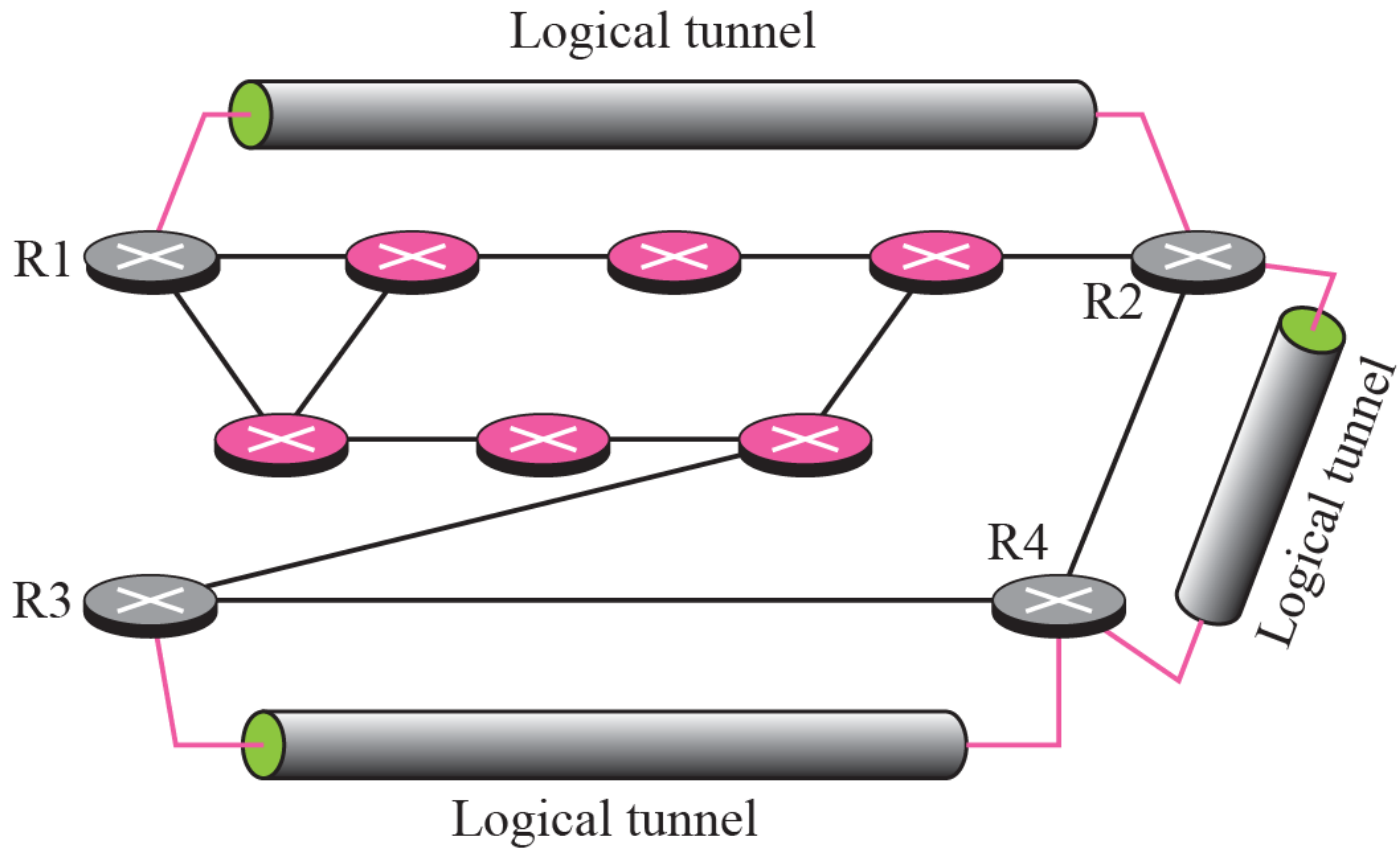


Figure 12.29 *MBONE*

